

Ensemble disagreement is miscalibrated as a scale, by a factor that grows with rollout depth: an independent reproduction of a released robotic world model

Anonymous authors

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Abstract

We rebuild the proprioceptive dynamics model of the *Robotic World Model* (arXiv:2501.10100v1) and its uncertainty-aware follow-up (arXiv:2504.16680v1) from scratch on CPU, checked against the released reference at gradient level. **The trust metric the method penalises with is miscalibrated as a scale, by a factor that grows with depth.** Ensemble disagreement is smaller than realised error by $8.3\times$ at $h = 1$ and $33.4\times$ at $h = 100$, where its imagination rollouts run. **That is not accumulated rollout error:** at one step nothing has accumulated and the input is the true state, yet it is already an order of magnitude out. **Nor is it a units problem a tuned coefficient absorbs.** The penalty enters through a scalar weight, so a uniformly wrong scale would simply rescale it — but the best constant rescale, an upper bound because it is fitted and scored on the same data, grows by a factor of 4 across the rollout. The σ the method computes and discards is worse still, and we derive why: the implemented objective’s optimum is $\sigma = 0$. **The base paper’s central training claim reproduces, and the advantage grows with horizon** — $4.61\times$ on the reference’s own relative-L1 error at $h = 368$, under a rule committed to git before the runs. **At one step it reverses:** under a second rule, a shorter evaluation unit resolves a gap the 400-step unit left spanning zero, and teacher forcing wins. **The released evaluation understates its own checkpoint:** it pairs each state with the previous step’s action, overstating that checkpoint’s nRMSE at $h = 368$ by 75%. As a *ranking* it is far better, with limits: holding rollout and depth constant it still tracks error and beats the forecast step index everywhere, but the model’s own predicted step size comes close enough that this sample cannot separate them. One repair works: a per-horizon multiplier, fitted on one held-out episode and scored on the other, restores nominal coverage on every held-out cell. Every **measurement** here is substituted from a named artifact, by a build that fails on any numeral it cannot classify.

1 Introduction

A world model that reports its own uncertainty is more useful than one that does not, and the uncertainty-aware Robotic World Model reports one. This paper asks what that number means.

We came to the question sideways. Our aim was an ordinary reproduction: rebuild the proprioceptive dynamics model from scratch, check it against the released implementation, and see whether the paper’s central training claim holds. It does. But the same rebuild made a second question cheap to ask, because we had a from-scratch model, the released checkpoint, and a harness that could score both: *is the predicted σ calibrated?* Neither of the two the checkpoint emits is — the per-member σ by three to four orders of magnitude, the ensemble disagreement the method actually uses by one to two — and for the first of them the reason is structural rather than incidental.

This is a reproduction paper, and we mean the term in its stronger sense: the contribution is not that the numbers came out the same, but what systematically re-measuring the method reveals about where it is

robust and where it is not. §9 collects the lessons in a form a practitioner can apply without reading the rest. Three things distinguish the work from a re-run of the authors’ code.

We rebuilt rather than imported. The forward pass, the loss and the training step are written from scratch and then checked against the reference: outputs match bitwise, and losses and gradients match to 0.000e+00 across 7 loss terms and 106 parameter tensors before any training begins (Appendix A). A discrepancy found later is therefore a property of the method, not of our wiring.

Decision rules were committed before the data. The verdicts below were fixed in advance, in git, with timestamps a reader can check (§8, Figure 4). One of them returned "cannot be settled" and we report that too.

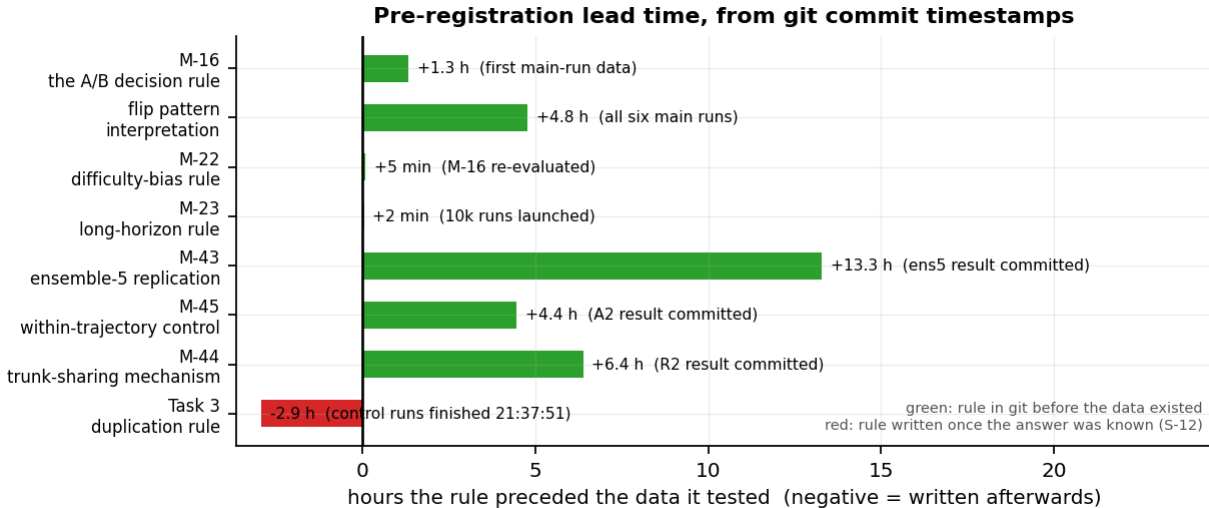


Figure 1: Pre-registration lead time for each decision rule, from git commit timestamps. Positive is a rule committed before the data that tested it existed; negative is a rule written afterwards. The one negative bar is the Task 3 duplication rule, retracted as a pre-registration in this paper.

We retract our own findings when they fail, on this project’s own evidence, and keep them in the record rather than deleting them; §8 and Appendix D give them in full. The count is in the contributions below, once.

Contributions.

- **A from-scratch reimplementaion verified at the gradient level.** Outputs match the released module bitwise; losses and gradients match to 0.000e+00 across 7 loss terms and 106 parameter tensors, before any training (Appendix A).
- **The base paper’s central training claim reproduces, and the advantage grows with forecast horizon:** a factor of $4.61\times$ on relative-L1 at $h = 368$ over 3 seeds, under a rule committed to git before the runs existed, rising monotonically to that from $2.58\times$ at $h = 100$ (§5).
- **At one step the comparison reverses, and we report it against our own arm.** A second pre-registered rule rebuilds the evaluation at a 33-row unit, giving 60 independent units where the 400-step unit gives 4. The gap becomes $-0.0194 [-0.0310, -0.0093]$ — it excludes zero in favour of **teacher forcing**. What the 400-step unit could only report as spanning zero is a real reversal (§5).
- **The first calibration measurement of either uncertainty output of this released checkpoint.** Lu et al. (2022) assess calibration for this family of penalties on models they train themselves (§2); we measure coverage against a nominal, on a checkpoint its authors deployed. Both outputs are overconfident by one to four orders of magnitude, with intervals over independent trajectories

at every horizon; and the aleatoric collapse is derived analytically from the implemented objective rather than observed (§6.2, §6.3).

- **A candidate mechanism for the epistemic failure, from source.** The five members share one trunk, one recurrent state and 89.15% of each member’s parameters, so their spread cannot express uncertainty the trunk does not already carry (§6.4).
- **The ranking claim tested against free baselines neither original ran,** and against six controls, the last of which removes trajectory difficulty rather than forecast depth and is the only one that isolates within-rollout information. Of the 2 baselines added here, disagreement beats 1: the model’s own predicted step size ranks error at +0.4697 against disagreement’s +0.6053, and the margin between them is below what this sample can resolve (§6.7).
- **The mechanism tested rather than asserted, under a rule committed before the runs.** An ensemble of 5 independently-initialised full models, sharing nothing, is $2.03\times$ better calibrated than the shared-trunk arms against a pre-registered minimum detectable effect of $1.45\times$. The decomposition says what that is made of: σ larger by $1.65\times$, 71% of the improvement at $h = 100$, reversing to 57% from accuracy at $h = 368$ (§6.10).
- **A working repair:** one multiplier per horizon, fitted on one held-out episode and scored on the other, restores nominal coverage where a global multiplier does not (§6.8).
- **The released evaluation is misaligned by one step, and the checkpoint is materially better than its own evaluation reports.** Training pairs states and actions index-for-index; evaluation feeds the action from $t-1$. Scored causally, nRMSE at $h = 368$ falls from 1.3228 to 0.7572 — the released harness overstates its own model’s error by 75% (§7.2). This is the finding most immediately useful to anyone using that repository, and it costs one line to fix.
- **12 retractions of our own claims,** kept in the record with the evidence that withdrew them, and Appendix G’s table of every pre-registered rule with its lead time and its verdict (§8).

2 Related work

This paper measures whether an ensemble-disagreement penalty is calibrated, and separates its use as a *ranking* from its use as a *scale*. Both questions have a literature, and one paper asks almost exactly ours four years earlier. Stating that plainly is not a concession: it tells a reader which part of what follows is a new measurement and which part is a new measurement *of a new object*.

The direct precedent. Lu, Ball, Parker-Holder, Osborne and Roberts (*Revisiting Design Choices in Offline Model-Based Reinforcement Learning*, ICLR 2022) compare uncertainty heuristics in offline model-based RL under protocols built, in their words, to "capture the specific covariate shift induced by model-based RL", explicitly in order to assess calibration. They report Spearman rank and Pearson bivariate correlation against true model error **separately**, and observe that "despite the similar rank correlations ρ , the bivariate correlations r can vary considerably" — that is, a configuration can preserve ordering while changing the relationship between the penalty’s magnitude and the error’s. That is our ranking-versus-scale distinction, on the same family of penalties.

Their results also bear on ours directly, and in the same direction. They find the ensemble standard deviation — the exact quantity `system_dynamics.py:126` computes and `envs/base.py:166` applies — to correlate with model error *better* than the penalties of MOPO and MOREL, being "strikingly similar" to the latter but better behaved. Our §6.7 finds the same quantity to be a usable ranking signal. Two independent measurements, on different systems, agreeing.

So what is left for us. Lu et al. ask their question of simulated benchmarks, of models they train themselves, and they do not ask whether the penalty is a calibrated interval — rank and bivariate correlation are both ordering-and-shape statistics, not coverage. We ask it of a **released checkpoint** from a pipeline its authors deployed on hardware, where the interval reading is consequential; we measure **coverage** against a nominal,

not only correlation; and we do it with a from-scratch reimplementaion verified against the reference at the gradient level, so that a discrepancy is a property of the method rather than of our wiring. The contribution is not the idea of checking. It is the object checked, the quantity measured, and the standard of verification.

Where the parameterisation comes from, and why it matters more than one repository. The bounded $\log\text{-}\sigma$ head that §6.3 shows has its optimum at $\sigma = 0$ is not this codebase’s invention. It is inherited, line for line, from the probabilistic ensembles of Chua, Calandra, McAllister and Levine (PETS, NeurIPS 2018). PETS’s Appendix A.1 gives

```
logvar = max_logvar - softplus(max_logvar - logvar)
logvar = min_logvar + softplus(logvar - min_logvar)
```

and `architectures/mlp.py:92-93` is those two lines in log-standard-deviation rather than log-variance, with `system_dynamics.py:302` supplying PETS’s regulariser on the bounds.

What is **not** inherited is the objective. PETS uses "the negative log prediction probability as our loss function", and it is that likelihood’s $\log\text{-}\sigma$ term which opposes $\sigma \rightarrow 0$. `system_dynamics.py:283` substitutes squared error on a reparameterised sample, which has no such term. So the collapse §6.3 derives follows from the *substitution*, not from the parameterisation — and that makes §6.3 larger than one repository: **any descendant of this lineage that replaced the likelihood with a sampled squared error inherits the same optimum**. We state that as a hypothesis and mark it clearly: we have not tested any other descendant, and testing one is out of scope here (§11).

The method being reproduced sits in a well-populated family. MOPO (Yu, Thomas, Yu, Ermon, Zou, Levine, Finn and Ma, NeurIPS 2020) penalises the reward by an ensemble uncertainty estimate to solve a pessimistic MDP; MOREL (Kidambi, Rajeswaran, Netrapalli and Joachims, NeurIPS 2020) builds an unknown-state detector from pairwise ensemble disagreement instead; and both branch short model rollouts from real states in the manner of MBPO (Janner, Fu, Zhang and Levine, NeurIPS 2019). The follow-up we reproduce adapts MOPO’s penalty into MBPO’s loop, which is what "MOPO-PPO" names. The rollout-length-versus-model-error trade MBPO introduces is the one the follow-up’s 100-step imagination horizon sits inside (§6.2, and X-13 in the ledger).

What "ensemble" is supposed to mean. Deep ensembles (Lakshminarayanan, Pritzel and Blundell, NeurIPS 2017) are several networks trained from *different random initialisations and different data orderings*, and their spread is the uncertainty estimate. That definition is the reference point for §6.4: the released checkpoint’s five members share one GRU trunk, one recurrent hidden state, and 89.15% of each member’s state-prediction parameters. Whatever that spread measures, it is not what a deep ensemble measures, and §6.4 sets out what follows.

Three sources of uncertainty, and the one nobody here estimates. Abbas, Sokota, Talvitie and White (ICML 2020) separate predictive uncertainty in model-based RL into aleatoric noise, parameter uncertainty, and **model inadequacy**, and observe that selective-planning work attends almost entirely to the second. The checkpoint we measure emits an aleatoric term and a parameter-uncertainty term, discards the first before use (§6.1), and has no estimate of the third at all. Model inadequacy is precisely the component that compounds with rollout depth, which is the shape §6.9 reports: σ flat while error grows.

Why miscalibration is the expected finding rather than a surprising one. Modern networks are systematically miscalibrated (Guo, Pleiss, Sun and Weinberger, ICML 2017), and calibration degrades further under covariate shift, worsening with distance from the training distribution (Ovadia, Fertig, Ren, Nado, Sculley, Nowozin, Dillon, Lakshminarayanan and Snoek, NeurIPS 2019). An autoregressive rollout manufactures its own covariate shift, increasing with depth, so horizon-dependent calibration failure is the shape one should expect. Our contribution on this axis is the *magnitude* and the *mechanism*, not the direction. Finally, §6.8’s per-horizon multiplier is a coarse instance of calibrated regression (Kuleshov, Fenner and Ermon, ICML 2018): a post-hoc map fitted on held-out data. We present it as an application of that idea to a horizon index, not as a new one.

The closest published work to our one constructive result. Malik, Kuleshov, Song, Nemer, Seymour and Ermon (ICML 2019) recalibrate a dynamics model’s uncertainty inside model-based RL, and argue that

"good uncertainties must be calibrated" rather than merely well ranked — which is the distinction §6.7 and §6.2 draw between what the released checkpoint's disagreement does and does not do. §6.8 is a horizon-indexed instance of that idea, and §6.8 says what is new relative to it rather than leaving a reader to work it out: the conditioning variable is the forecast horizon, and a single global multiplier **fails** where a per-horizon one works. That distinction is not decoration — an open-loop rollout's error accumulates with depth, so a horizon-blind recalibration cannot follow it, and ours is the measurement showing it does not.

§6.4's mechanism is known, and we say so. That heads sharing a trunk under-report disagreement relative to independently initialised networks is established: Lee, Purushwalkam, Cogswell, Crandall and Batra (arXiv:1511.06314) treat ensemble diversity as something to be engineered rather than assumed; Fort, Hu and Lakshminarayanan (arXiv:1912.02757) show that what independent initialisation buys is decorrelation that subspace methods do not match; and BatchEnsemble (Wen, Tran and Ba, ICLR 2020) and MIMO (Havasi, Jenatton, Fort, Liu, Snoek, Lakshminarayanan, Dai and Tran, ICLR 2021) share deliberately and state what they trade away. §6.4 is not the discovery of that effect. **What is ours is finding it in a released robotics checkpoint that its authors deployed on hardware, with the sharing quantified at 89.15% of each member and the cost measured at $2.03\times$ (§6.10).** The problem is not sharing; it is sharing and then reading the spread as though the members were independent.

And the objective in §6.3 has a neighbour. Seitzer, Tavakoli, Antic and Martius (ICLR 2022) identify failure modes of heteroscedastic σ heads trained by maximising log-likelihood. The released model's state loss is not a log-likelihood at all — a *sample* enters a squared error — so the failure §6.3 derives is more basic than the ones they characterise, and it does not depend on the optimiser. §6.3 gives the derivation and demonstrates it against known noise.

Every entry above was checked against the paper itself — title, full author list, venue and year from the arXiv record, and for any sentence we attribute, the sentence matched verbatim against the paper's own text. 16 of 16 entries verified, 17 of 17 attributed fragments verbatim (results/t1_bibliography_verified.json). No entry was added that was not verified.

3 Setup

Data. The released dataset is 10,000 rows of ANYmal D proprioceptive state and policy actions at 50 Hz. It is not one recording: it is ten concatenated 20-second episodes, and its termination column is identically zero, so nothing in the file marks the boundaries.

The segments are not all the same length, and the arithmetic below does not work if one assumes they are. The first episode is 999 rows and the other nine are 1,000 each, with a reset at rows 999, 1,999 ... 9,999 and 1 orphan row at the end of the file that begins an eleventh episode and ends immediately: $999 + 9 \times 1,000 + 1$. The boundaries are recovered from the data rather than assumed — at every reset row the twelve joint velocities, the four HAA joint positions and all twelve actions are exactly zero, and no other row has that fingerprint. Ten equal segments of 1,000 would give one fewer crossing window and one more usable one, so a reader checking the counts below against the obvious reading would find them off by one and be right to. `results/step0_regimes.json` re-derives the crossing count a second time from the segment lengths alone, and the two agree.

A window is 40 rows — 32 of history and 8 of forecast — so the reference window builder marks all 9,961 windows valid, including 352 that splice one episode's end onto the next one's start. The usable, episode-respecting count is 9,609 — 10,000 rows, less 39 that cannot start a full window, less 352 that cross a boundary. The contamination rate is 3.53%.

Model. A GRU-based ensemble predicting the next proprioceptive state, with a mean head and a bounded log- σ head, plus auxiliary heads for contact and termination. The paper describes two loss terms; the implementation has 7.

Evaluation. Two arenas, held separate throughout: *out-of-sample*, the two episodes withheld from training, and *in-sample*, the eight used for it. We report both, because the released evaluation draws its trajectories from training data and the distinction is invisible in the original.

Effective sample size. Trajectory count is not sample size. Two 400-step trajectories whose spans overlap are not independent evidence, and the out-of-sample arena contains only 4 mutually non-overlapping 400-step trajectories. Every interval in this paper is a bootstrap over independent trajectories, and every table reports that count.

3.1 Metrics

Every metric below is stated as implemented, with the `file:line` of the implementation; each citation is read back and checked against its own source text on every build (20 of them). This section exists because two metrics in this project once disagreed in *direction* at $h = 1$, and because a choice between two aggregations of the same metric once inverted a comparison against the released model. A reader who cannot see the denominator cannot check the headline.

Relative-L1 is the reference’s own metric, reproduced verbatim in behaviour (`model_training.py:203`) so that our numbers are comparable to the upstream’s printed one. On config-normalised states, per forecast step,

$$r_t = \frac{\sum_{d=1}^{45} |\hat{s}_{t,d} - s_{t,d}|}{\sum_{d=1}^{45} |s_{t,d}|}$$

and the reported figure is the flat mean over trajectories and steps,

$$e = \frac{1}{B(T - t_0)} \sum_{b=1}^B \sum_{t=t_0+1}^T r_t^{(b)}$$

with $t_0 = \text{history_horizon} = 32$: the first 32 steps are teacher-forced and excluded. The denominator is recomputed at every step and is a 45-term sum in normalised space, so it can pass through zero — which is why this metric goes non-finite on low-dimensional state groups, and why a second one exists.

Normalised RMSE fixes the denominator once, over the training episodes only:

$$\text{nRMSE}_t = \frac{\sqrt{\frac{1}{45} \sum_{d=1}^{45} \text{MSE}_{t,d}}}{\frac{1}{45} \sum_{d=1}^{45} \sigma_d^{\text{tr}}} \quad \text{with} \quad \text{MSE}_{t,d} = \frac{1}{B} \sum_{b=1}^B (\hat{s}_{t,d}^{(b)} - s_{t,d}^{(b)})^2$$

where the scale constant is

$$\sigma_d^{\text{tr}} = \text{sd}(\{ \tilde{s}_{i,d} : \text{episode}(i) \in \mathcal{E}_{\text{train}} \})$$

computed once, stored in `results/step4_0a_results.json`, never recomputed per step and never derived from held-out data. A value of 1.0 means no better than predicting the training mean. **The aggregation matters and is form 1:** pool the per-dimension mean squared errors, then divide — a ratio of means. The alternative, a mean of per-dimension ratios, gives whichever dimension has the smallest scale unbounded leverage, and the choice between the two once inverted a published-model comparison in this project. Appendix I gives both forms and the inversion.

Coverage at $\pm k\sigma$ is the fraction of scalar (trajectory, forecast step, state dimension) triples whose absolute realised error falls within k times the σ predicted for that same triple:

$$\text{cov}_{\pm k\sigma}(h) = \frac{1}{B h 45} \sum_{b=1}^B \sum_{t=t_0+1}^{t_0+h} \sum_{d=1}^{45} \mathbf{1} \left[\frac{|\hat{s}_{t,d}^{(b)} - s_{t,d}^{(b)}|}{\sigma_{t,d}^{(b)}} \leq k \right]$$

Three things a reader needs and the prose did not previously give. It is pooled over all three axes with equal weight per triple. It is **cumulative** over steps 1..h — coverage "at h" averages the whole rollout up to h and is not the value at step h, and the same convention governs every horizon-indexed quantity in this paper. And because the statistic is built from an *absolute* error, $z \leq k$ is the two-sided event, so the calibrated targets are $\text{erf}(k/\sqrt{2})$: **68.27%** at $\pm 1\sigma$ and **95.45%** at $\pm 2\sigma$. **That nominal is checked rather than assumed.** Rescaling each model's σ by the single constant that makes $\text{mean}|\text{error}| / \text{mean } \sigma$ equal a calibrated Gaussian's 0.7979 lands coverage at or above 68.27% in every one of 48 model $\times$ horizon cells, with 0 below it (M-65). The shortfalls reported below are therefore a property of the scale and not of the tail.

The overconfidence factor is how many times larger the typical realised error is than the typical predicted σ :

$$\rho(h) = \frac{\text{mean}_{b,t \leq h,d} |\hat{s}_{t,d}^{(b)} - s_{t,d}^{(b)}|}{\text{mean}_{b,t \leq h,d} \sigma_{t,d}^{(b)}}$$

It too is a **ratio of means**, not a mean of ratios — the latter is unbounded whenever a single σ approaches zero, which is exactly the regime §6.3 puts these models in. One caution on reading it: $\rho = 1$ is *not* calibration. A calibrated Gaussian has $\text{mean}|\text{error}| / \sigma = \sqrt{2/\pi} = 0.7979$. ρ is reported as a magnitude of miscalibration and coverage is the calibrated reading, which is why both appear everywhere.

Which metric each headline uses. The A/B training claim (§5) is relative-L1, because the claim is about reproducing the upstream's comparison and that is the upstream's metric. The calibration claims (§6.2) are the overconfidence factor and coverage, because neither error metric involves σ at all. The ranking claims (§6.7) are Pearson correlations between the applied scalar penalty and total absolute error, because a ranking claim is about order rather than scale. Every headline number in the abstract names its metric, including §7.2's alignment defect, which is nRMSE and is horizon-specific: the same comparison overstates by 75% at $h = 368$ and by 9.5% on relative-L1, so quoting it without both scopes invites a reader to check it against the wrong number.

Horizons. Curves are reported at $h \in \{1, 8, 32, 100, 128, 368\}$. Two of those are load-bearing and the rest are landmarks. $h = 100$ is the method's own imagination rollout length — the horizon over which the uncertainty-penalised policy loop actually runs this model (arXiv:2504.16680 Table S9 in v1, Table S11 in v3; the value is unchanged between them, and v3 states it in prose as well). $h = 368$ is the upstream's open-loop diagnostic length: `len_eval_trajectory` = 400 minus the 32-step teacher-forced prefix, which is the curve the follow-up plots as its uncertainty figure. It is 3.68 $\times$ the method's own rollout length and it is not a deployment horizon; earlier drafts of this paper called it one, and that label is withdrawn. Both are kept in every table, because $h = 368$ is what makes our numbers comparable to the original's *figure* while $h = 100$ makes them comparable to the original's *method*.

4 What the original papers claim, and which claims we test

A reproduction that does not say what it left alone invites the reader to assume it tested everything. It did not.

We tested four claims and left eight untested. The four are the base paper's autoregressive -versus-teacher-forcing comparison and its claim that teacher forcing generalises poorly (§5), and the follow-up's two claims about what its uncertainty outputs report (§6). Of the eight we did not test, **six need a simulator we do not have**: zero-shot transfer, the sample-efficiency result, the comparisons against SHAC and Dreamer, generality across robot morphologies, offline MBRL on real robots and the core claim that penalising rewards by disagreement improves the learned policy. **Five of those six are claims about policy learning or hardware** — every one but generality across robot morphologies, which needs a simulator and recorded data from other robots but is a claim about the model rather than about a policy, and does not belong in that bucket however convenient it is to put it there. This work trains no policy at all and runs on two CPU cores. **The remaining two need none of that and we still did not run them**: the M/N configuration

sweep and the MLP/RSSM/transformer baseline comparison are within reach of the CPU budget this project already spent, and Appendix E prices both. They are unrun for want of time, not for want of hardware.

Both counts and both lists above are generated from a classification tag carried in Appendix F’s verdict column, so the enumeration cannot disagree with the count beside it. It did: an earlier draft of *this* sentence — itself the replacement for retracted claim S-17, which was a count defect in this same place — said six and then named Five, and called all of them claims about policy learning or hardware when one of them is not. S-17 withdrew a universal quantifier here; its replacement got the arithmetic wrong instead, which is the worse failure of the two because the sentence had just been rewritten under scrutiny. §11 states what that bounds, and Appendix E sets out what testing them would take.

For all 4 of the claims we did test, the original reports no quantitative figure. Each is asserted qualitatively and shown in a plot; none is given a number in text, caption or table. So our $4.61\times$ at $h = 368$ is neither a confirmation of a published figure nor a contradiction of one — it is the first figure attached to the claim, and the same is true of the follow-up’s “strong correlation” between disagreement and error, for which §6.7 supplies the first coefficient. Where a magnitude is legible only from a plotted curve we say so rather than estimating it from the axis.

Appendix F gives the full table, claim by claim, with what the original states, where it states it, and our verdict.

5 The base paper’s central claim reproduces

Claim under test. Training the dynamics model on its own autoregressive rollouts beats training it with teacher forcing, at long forecast horizons.

Which horizon, stated before the result. The rule below is anchored at $h = 368$, which §3.1 identifies as the upstream’s **open-loop diagnostic** length and explicitly not a deployment horizon. That is the horizon the rule was committed over and the horizon its verdict is returned at; we do not re-anchor a discharged rule. But the paper’s own deployment horizon is $h = 100$, so the same comparison is reported there too, below, and the two differ in size.

Rule, committed in advance (commit efc35b8, and it names conditions rather than outcomes). Three conditions, all required: the out-of-sample gap at $h = 368$ excludes zero under a bootstrap over independent trajectories; the sign is consistent across episodes; and the effect survives at 10,000 iterations rather than only at the paper’s 2,500.

Result. Every condition holds. We give the evidence in order of how little it depends on the small held-out sample.

The sign test, which does not depend on n . At $h = 368$ the per-episode gap favours autoregressive training on **10 of 10** episodes — an exact two-sided binomial test, $p = \mathbf{0.0020}$. At $h = 100$ it is **10 of 10**, $p = \mathbf{0.0020}$, so the count the abstract leans on is not an artifact of the longest horizon; at $h = 1$ it is 3 of 10, which is the same story the interval tells. This is one test on ten paired episodes, it uses no bootstrap, and no multiplicity correction touches it. Unlike the per-dimension counts in §6, episodes are genuinely separable units, so a binomial null is admissible here.

Its scope needs stating plainly, because the arena labels invite a stronger reading than it supports. 8 of the 10 episodes are training data for *both* arms. The test is therefore a valid **paired** comparison — the two arms saw identical data, so any episode-level difference is attributable to the training rule and not to what either model had memorised — but it is not ten out-of-sample episodes, and it does not measure generalisation. The out-of-sample effect size below carries that burden, on 4 independent trajectories.

The in-sample arena, where the sample is larger. The same comparison on the eight training episodes has 16 independent 400-step trajectories against the held-out arena’s 4 — $4\times$ more — and gives the same direction at every horizon and checkpoint.

The out-of-sample effect size, at every horizon rather than one. At $h = 368$, the horizon M-23 is stated over, autoregressive training reaches 0.3582 ± 0.0283 against teacher forcing's 1.6497 ± 0.2858 (standard deviation over seeds, $\text{ddof}=1$) — a factor of $4.61\times$. At $h = 100$, the method's own imagination rollout length and the horizon everything in §6 is anchored to, the same three seeds give $2.58\times$.

Quoting one of those and not the other would be a choice, so we report the curve (Figure 6). Same rollouts, same 3 seeds, same held-out arena, $n_{\text{independent}} = 4$, with a cluster bootstrap over whole trajectories:

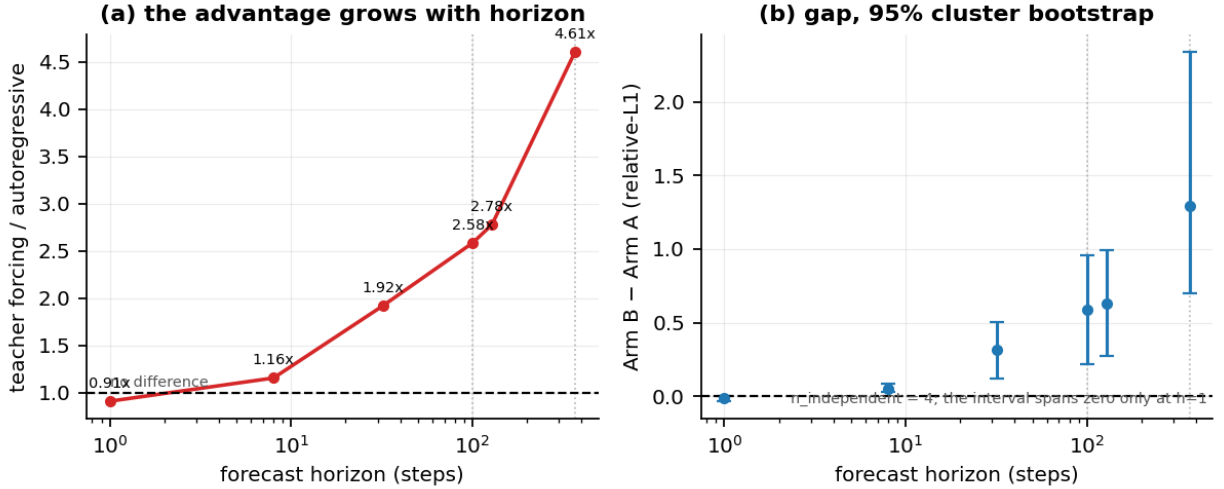


Figure 2: The autoregressive-versus-teacher-forcing advantage as a function of forecast horizon, out-of-sample over three seeds. (a) the ratio, which grows monotonically with depth: $h = 368$ is the end of a trend rather than a selected point, and the method's own rollout length of $h = 100$ sits partway along it. (b) the same comparison as a gap with its 95% cluster-bootstrap interval over whole trajectories; the interval spans zero only at $h = 1$, where teacher forcing is ahead – a lead a shorter evaluation unit resolves as real, not nominal (§5, M-64). Only the $h = 368$ figure is pre-registered (M-23); the rest were computed after the data existed.

h	autoregressive	teacher forcing	ratio	gap [95% CI]	excludes 0	hold-last floor	A vs floor	B vs floor	episodes A leads
1	0.1447 ± 0.0023	0.1319 ± 0.0172	0.91x	-0.0128 [-0.0329, +0.0013]	no	0.0796	0.5x	1.66x	3/10
8	0.3087 ± 0.0318	0.3572 ± 0.0088	1.16x	+0.0485 [+0.0271, +0.0844]	yes	0.3298	1.1x	1.08x	10/10
32	0.3415 ± 0.0491	0.6555 ± 0.0699	1.92x	+0.3140 [+0.1233, +0.5047]	yes	0.5950	1.7x	1.10x	10/10
100	0.3700 ± 0.0290	0.9561 ± 0.0211	2.58x	+0.5861 [+0.2174, +0.9547]	yes	0.7558	2.0x	1.27x	10/10
128	0.3558 ± 0.0231	0.9881 ± 0.0376	2.78x	+0.6324 [+0.2716, +0.9931]	yes	0.7999	2.2x	1.24x	10/10
368 (M-23)	0.3582 ± 0.0283	1.6497 ± 0.2858	4.61x	+1.2915 [+0.7004, +2.3390]	yes	0.9930	2.8x	1.66x	10/10

The advantage does grow monotonically with forecast depth. Over 400-step trajectories the gap excludes zero at 5 of 6 horizons and spans it at $h=1$. That is the reading a single figure cannot give: $h = 368$ is the end of a trend rather than a point we picked, $h = 100$ sits partway along it, and the claim is weakest exactly where the model is trained.

At $h = 1$ that is weaker than the data now supports, and the correction runs against us. The row above rests on 4 independent 400-step trajectories. A 400-step unit is required only by the longest horizon, so under a rule committed before the index was built (M-64) we rebuilt it at 33 rows — 32 of history and one forecast step, non-overlapping within an episode — which yields 60 units on the same two episodes. The gap is -0.0194 [-0.0310, -0.0093]: it **excludes zero, in favour of teacher forcing** (0.85x). Both readings are true at their own unit and both are reported — the 400-step row is what M-23 was discharged over and stays as the table, and the short unit is what resolves the sign. So the $h = 1$ result is not "no difference" but **autoregressive training is worse at one step**, which is the direction the sign test and the hold-last floor already pointed.

Only the $h = 368$ row is pre-registered. M-23 was committed at that horizon, before the runs, and its verdict stands as returned. Every other row was computed after the data existed, so by this paper’s own standard (§8) it is not a pre-registration and carries none of the weight one would — the same treatment §6.7 gives the expectation we held about the counter-baseline. Nothing in the table discharges or re-opens M-23; the rule’s anchor being the diagnostic horizon rather than the deployment one is recorded as M-46.

Two things in that table were not visible from $h = 368$ alone, and one of them cuts against us. Teacher forcing is worse than the hold-last floor at every horizon, so §5’s sharpest line is not an artifact of the longest horizon — at $h = 100$ it is still $1.27\times$ worse than assuming nothing changes. But **at $h = 1$ the floor beats both arms:** it scores 0.0796 against autoregressive training’s 0.1447, the only horizon where a trained model loses to predicting no change at all. At 50 Hz one step is 20 ms and the state barely moves, so that is what one should expect; it is stated because §5 quoted the $h = 368$ margin over the floor with no indication that it does not hold everywhere.

Seed spread is not symmetric between the arms and that is worth stating: Arm A ranges 0.3341–0.3894 across seeds (7.9% relative), Arm B 1.4241–1.9710 (17.3%). Teacher forcing is more than twice as variable across seeds as autoregressive training at this horizon, so a single-seed comparison of these two arms is unreliable in a way a reader should know about. An earlier draft of this paper quoted the single-seed figures 0.3509 and 1.5540; those came from the seed that happened to be favourable to Arm A and unfavourable to Arm B, and the three-seed ratio at $h = 368$ is $4.61\times$ rather than $4.4\times$.

For a single seed the bootstrap over trajectories gives 95% interval [0.56, 2.05] on $n = 4$ independent trajectories. **That interval should not be read as an ordinary one:** four trajectories admit 256 distinct resamples, so any bootstrap tail is quantised to steps of 0.39%, and the interval is coarse by construction. It is offered as corroboration of the sign test, not as the primary evidence.

So here are the four numbers the interval is computed from, which are more informative than the interval and cost nothing to print. Three seeds pooled, the per-trajectory gap (Arm B minus Arm A) at $h = 368$: **+2.8705, +0.8949, +0.7445, +0.6562.** All 4 of 4 are positive, which is the sign test; but one trajectory carries +2.8705 against a smallest of +0.6562, and no interval on four units shows a reader that. At $h = 100$ the four are +1.0925, +0.8170, +0.2800, +0.1549.

The same is done for the other comparison this paper makes at $n_{\text{independent}} = 4$: §6.10 and §11’s paired contrasts carry their four per-trajectory ratios and coverage differences in `results/r2_independent_ensemble.json` and `results/m49_capacity_matched.json`. **It is not done everywhere.** §6.2’s calibration tables report intervals at $n_{\text{independent}} = 4$ in every cell, and printing four values in each would be sixty-four numbers in one table; those intervals are quantised at the resolution §3 states and should be read as coarse. An earlier version of this paragraph claimed the underlying values were in the artifact for *every* such interval. They were not — only §5’s were — and the claim is narrowed to what is true.

Against a baseline, because neither number means anything without one. The hold-last floor — predicting that nothing changes — scores **0.9930** in the same $h = 368$ cell, and autoregressive training beats it by **$2.8\times$** there and by $2.0\times$ at $h = 100$. **Teacher forcing is $1.66\times$ worse than assuming nothing changes at all** at $h = 368$, and $1.27\times$ worse at $h = 100$: the arm that reaches a lower training loss ends up predicting the future worse than a model that makes no prediction, at every horizon we measured. That is the sharper statement of what exposure bias costs here. **The floor is not a weak baseline everywhere,** and the table above says where it is not: at $h=1$ it beats the autoregressive arm as well.

What does not hold, and we say so. At $h = 8$ — the horizon the model is trained on — the same comparison out-of-sample gives a gap of 0.008 whose interval includes zero. The advantage is a long-horizon phenomenon. An earlier rule of ours, anchored at $h = 8$, returned "cannot be settled"; anchoring a rule to the horizon the claim is actually about was a correction we had to make in advance of the runs, not after them (§8).

The pattern is consistent across the design. Under the correct cluster bootstrap, the out-of-sample gap excludes zero in **4 of 4** long-horizon cells — both trajectory lengths crossed with both checkpoints — and

in **0 of 4** at $h = 8$. These figures are relative-L1; the nRMSE aggregation is reported separately and does not change the direction.

Multiplicity. Those 4 cells sit in a family of 8 out-of-sample comparisons, so we state the correction rather than leaving it to a reader. All 4 of 4 still exclude zero at a Bonferroni level of $0.05/8$, and Holm–Bonferroni rejects **4 of 4**. The sign test above is unaffected either way.

5.1 The data budget, which is the one part of the sample-efficiency claim we can measure

The base paper’s headline is a sample-efficiency result: policies transfer to hardware from 6,000,000 state transitions of world-model pretraining against $\sim 250\text{M}$ for the model-free baseline (Table I). We cannot test it — it is a claim about policy learning and hardware. But its *world-model* half is a claim about a quantity we can count exactly, and ours is directly comparable.

Our arms consume 7,991 distinct state transitions. That is the 7,999 rows of the eight training episodes less one per episode boundary (8 of them), a transition being a consecutive pair of rows inside one episode. It is deliberately not the 7,687 training windows, which overlap almost completely — consecutive 40-row windows start one row apart — nor the 640,000 window draws a run makes, which resample the same data with replacement. Against the reference’s 6,000,000, that is **$751\times$ less data, 0.133% of its world-model budget**.

A dynamics model trained on 0.133% of the reference’s data still reproduces the autoregressive-versus-teacher-forcing result — $4.61\times$ at $h = 368$ and $2.58\times$ at $h = 100$ — and still beats the hold-last floor, by $2.8\times$ and $2.0\times$ at those two horizons. That is what this paper can contribute to the sample-efficiency question without training a policy.

Three limits, in the same breath. It is not a reproduction of the 6,000,000-against-250M comparison, which is about policy learning and which we do not touch. It says nothing about whether a policy trained inside our model would transfer to hardware, or anywhere. And our model is evaluated on the same narrow distribution it trained on — one robot, one gait, one terrain, velocity commands from a single bounded box — where the reference’s 6,000,000 transitions span considerably more. A smaller data budget buys less than it appears to when the evaluation distribution shrinks with it.

6 Neither of the checkpoint’s uncertainty outputs is usable as an interval

6.1 Which quantity the method actually uses

The released checkpoint emits **two** uncertainty quantities, and the method consumes only one of them. This has to be settled before any calibration number means anything.

`system_dynamics.py:125` computes an **aleatoric** term, the mean over ensemble members of each member’s predicted σ . `system_dynamics.py:126` computes an **epistemic** term, the standard deviation across the members’ mean predictions. In `envs/base.py:142` the aleatoric term is bound to a local variable that is never read again; the epistemic term is stored, returned to the policy loop at `:158`, and applied at `:166` as a reward penalty with weight -1.0 .

The paper agrees with its code. arXiv:2504.16680v1 Eq. 4 defines the penalised quantity as — the numbering is unchanged in the current v3 — $u = \text{Var}_b[\mu_b]$, the variance across ensemble members, and Eq. 5 applies it as $\tilde{r} = r - \lambda u$. The per-member predicted variance enters the training objective and nothing downstream.

Eq. 4 specifies a **variance** while `system_dynamics.py:126` computes a **standard deviation**, which with $\lambda = 1$ differ by a square. We asked, and the first author confirms the code is operative: the penalty is applied to the standard deviation as intended, and Eq. 4 is "more of a high-level explanation" (personal communication, 21 August 2026; the exchange is reproduced in full, anonymised, in the supplementary material as `SUPPLEMENTARY_CORRESPONDENCE.md`, so these quotations are checkable rather than asserted). We measure the code’s quantity throughout, which is now known to be the intended one.

The same correspondence confirms the discard directly: "the aleatoric term is not used in downstream training. It is reported in Fig. 3 (right) as an analysis of the model behavior." (The figure number is the first author's and follows arXiv:2504.16680v1; in the current v3 that figure is Fig. 4. The crosswalk is in `results/original_paper_figures.json`.) So what follows is not an implementation slip being reported back to its authors — it is the intended design, and the aleatoric head exists to shape training and be inspected rather than to be consumed.

So the aleatoric head — the one the state loss and the bound loss shape, and the one §6.3 explains — is computed on every imagination step and discarded. We report both quantities below. Our own arms are ensemble size 1, where the epistemic term is identically zero by construction, so the epistemic measurement is possible only on the released checkpoint.

What the follow-up does and does not claim, stated before we measure anything. It does not claim its uncertainty is a calibrated interval. The follow-up's §5.1 claims the epistemic term "closely follows the trend of the prediction error" and that this "justifies its role as a trust metric", and of the aleatoric term it observes only that it "remains low, reflecting small stochasticity in the environment". Our measurement **supports the first claim** — the epistemic ordering is real and strong. What follows is therefore not a refutation of a calibration claim nobody made. The correspondence quoted in this section is quoted with the first author's permission, given on the record (§13). It is three things the papers do not address: that the aleatoric head is discarded before use, that neither quantity is usable as a scale, and that the low aleatoric value has a different cause than the one offered.

6.2 The measurement

For each model we compute the mean predicted σ , the mean absolute realised error, and the fraction of realised errors falling inside $\pm 1\sigma$. A calibrated Gaussian puts 68.27% inside $\pm 1\sigma$ (§3.1).

model	mean error / mean σ , whole 368-step rollout	coverage at $\pm 1\sigma$, h=1	coverage at $\pm 1\sigma$, h=100
faithful Arm A (sampled MSE)	52.2× [39.6, 69.7]	11.67% [7.96, 15.19]	2.17% [1.59, 2.75]
corrected Arm A (<code>gaussian_nll</code>)	10.9× [7.8, 14.7]	42.78% [24.44, 62.04]	9.8% [6.80, 12.81]
teacher-forced Arm B	315× [177, 509]	12.96% [7.04, 20.93]	1.22% [0.84, 1.61]
released checkpoint	7,878× [5,410, 9,934]	0.56% [0.00, 1.67]	0.08% [0.06, 0.11]

Every cell carries a 95% interval from a cluster bootstrap over whole trajectories, $n_independent = 4$; where three seeds contribute, seeds are pooled inside each draw rather than resampled, because seeds are not trajectories (§8). At $n_independent = 4$ the bootstrap has 256 distinct resamples and the intervals are quantised at that resolution.

All four rows are the held-out arena — the two episodes withheld from our own arms, $n_independent = 4$ — because that is the only arena on which our arms can be scored fairly. On the aleatoric head every model is overconfident by between one and four orders of magnitude (Figure 1); that is the quantity §6.1 shows the method discards.

A note on the released checkpoint's row, so the next table does not read as a contradiction. Its 7,878× is the whole 368-step rollout on those same 4 trajectories, for comparability with the three arms beside it. The released checkpoint trained on all ten episodes, so its own best-sampled figure is the 11,683× below — cumulative to $h = 100$, at $n_independent =$

1. **Both are correct and they differ in two ways at once: a different arena and a different horizon.** Neither is a held-out measurement *of the released checkpoint*, which has no held-out arena in this dataset.

The quantity the method does use is also uncalibrated. On the released 5-member checkpoint over all 10 episodes, $n_independent = 20$ non-overlapping 400-step trajectories. We use all ten rather than the held-out pair here because the released checkpoint trained on all ten, so restricting it to two buys no independence and costs four fifths of the sample — the same argument this paper makes about that

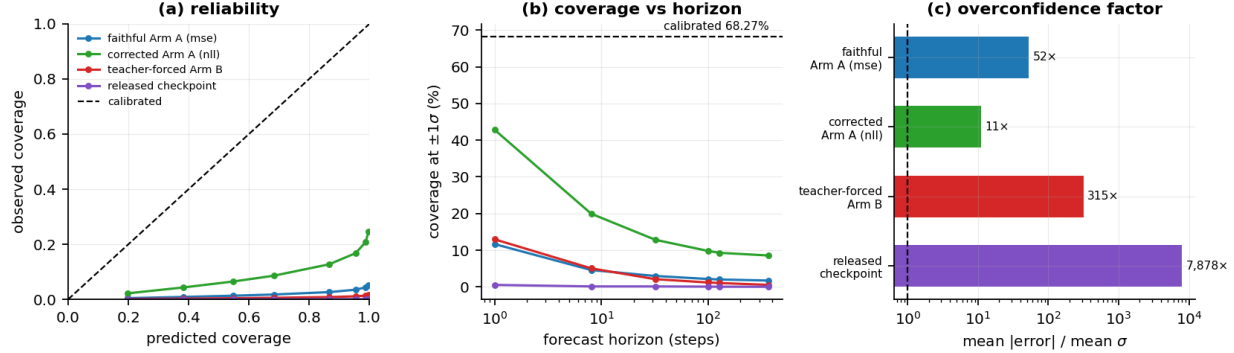


Figure 3: Calibration of all four models on the held-out arena. (a) reliability: observed against predicted coverage, with the calibrated diagonal. (b) coverage at $\pm 1\sigma$ against forecast horizon, log scale, against the 68.27% a calibrated Gaussian gives. Every curve sits far below the diagonal and falls further with horizon.

checkpoint elsewhere. The held-out-only version at $n_{\text{independent}} = 4$ is in the supplementary material (`results/task_b2_epistemic.json`). The epistemic column agrees in direction with this one at all 6 of 6 horizons; the aleatoric column agrees at 4 of 6 — it flips sign at $h=8$ and $h=100$, where both readings sit close enough to chance that the sign is not meaningful in either, and where the aleatoric σ is in any case three to four orders of magnitude too small for its ordering to be the interesting quantity. Where the two tables differ materially we say so.

h	aleatoric err/ σ [95% CI]	aleatoric $\pm 1\sigma$	epistemic err/ σ [95% CI]	epistemic $\pm 1\sigma$ [95% CI]	epistemic $\pm 2\sigma$	dims $r>0$	permutation P
1	$1,827\times$ [915, 3,110]	0.11%	$8.3\times$ [6.1, 9.7]	16.22% [13.44, 18.89]	30.11%	44/45	0.0056
8	$3,034\times$ [1,638, 4,849]	0.08%	$15.1\times$ [10.3, 19.2]	9.99% [8.60, 11.32]	19.76%	45/45	0.0069
32	$4,525\times$ [2,492, 7,196]	0.07%	$22.6\times$ [15.5, 28.7]	6.95% [5.99, 7.85]	13.68%	45/45	0.0344
100	$11,683\times$ [7,970, 15,846]	0.04%	$33.4\times$ [28.7, 39.0]	4.61% [4.12, 5.09]	9.27%	45/45	0.2769
128	$14,934\times$ [10,564, 19,700]	0.03%	$34.2\times$ [30.6, 38.2]	4.37% [3.96, 4.81]	8.75%	45/45	0.3762
368	$20,669\times$ [15,666, 25,688]	0.02%	$34.4\times$ [29.8, 40.3]	3.59% [3.27, 3.92]	7.19%	45/45	0.0804

Read the $h = 1$ row first. At one step the disagreement the method penalises rewards with is already $8.3\times$ [6.1, 9.7] smaller than realised error, with $\pm 1\sigma$ coverage of 16.22% [13.44, 18.89] against a calibrated 68.27%. The discarded per-member σ is $1,827\times$ out at $h = 1$ as well. Everything further down the table is deterioration from a starting point that is already broken.

Why that row and not the deep ones — the objection this section has to meet. A reviewer will say: a per-step predicted σ is a *conditional* quantity. It answers "given this input, how uncertain is the next state?", and in an open-loop rollout the input is the model's own previous output, which is wrong by an amount σ never claimed to describe. Comparing a conditional σ against *accumulated* rollout error is then comparing two different things, and the 368-step figure would be an artifact of that mismatch rather than a finding about the σ head. §6.9 answers a different objection — that the model is trained on 8 steps and cannot be expected to speak about 368 — and does not answer this one.

$h = 1$ answers it, and costs no new computation. At one step there is no accumulation, no compounding and no mismatch: the input *is* the true state, the prediction is one step ahead, and σ is being asked exactly the question it was trained to answer. At $h = 1$ it is out by $8.3\times$, and 16.22% of outcomes fall inside an interval that should hold 68.27%. Whatever compounding does at depth, it did not do that.

And that row is measured on data the checkpoint trained on. The released checkpoint trained on all ten episodes, and this arena is all ten — 10 of 10 of them (`results/insample_framing.json`). In-sample measurement biases *toward* better calibration, so the $8.3\times$ at $h = 1$ is if anything flattering. We stated the arena before and declined the inference; the inference is that the one-step figure is an upper bound on how well this checkpoint is calibrated.

So the horizon curve is not the claim; it is the shape of the deterioration, and the claim is the $h = 1$ row. We keep $h = 100$ because it is where the method actually deploys, and $h = 368$ because it is the upstream’s own diagnostic length — landmarks on a curve, not the evidence.

At $h = 100$, the method’s own imagination rollout length, epistemic is $349\times$ better than aleatoric and still wrong by $33.4\times$ [28.7, 39.0], with $\pm 1\sigma$ coverage of 4.61% [4.12, 5.09] where a calibrated Gaussian gives 68.27%. At one step it is $8.3\times$ out and $220\times$ better than aleatoric; at $h = 368$ the two-term ratio is $600\times$. **The gap between the two uncertainty terms is itself horizon-dependent, which is why each figure above names its horizon.** At the open-loop diagnostic horizon of $h = 368$ it is $34.4\times$ [29.8, 40.3] with 3.59% coverage — barely different, which is why the re-anchoring changes the reading and not the conclusion. **Total uncertainty, $\sqrt{\text{aleatoric}^2 + \text{epistemic}^2}$,** equals the epistemic value to four significant figures at every horizon, because the aleatoric term is too small to move it.

A constant scale error would not matter, and this one is not constant. The penalty enters as $r = r - \lambda u$, so a u uniformly c times too small is arithmetically identical to running with λ/c , and λ is tuned. The measurement answers this directly. Rescaling σ by the single constant that best calibrates it — fitted and scored on the same data, so it is an upper bound on what *any* constant achieves — needs 10.4 at $h = 1$ and 43.2 at $h = 368$, a factor of 4 across the rollout (M-65). No single λ is both of those. The mechanism is §6.9’s: σ is nearly flat while error grows by an order of magnitude, so the ratio runs $8.3\times$ at $h = 1$, $33.4\times$ at $h = 100$ and $34.4\times$ at $h = 368$. The penalty is applied at every step of a 100-step imagination rollout, so its *profile across the rollout* is wrong in a way no rescaling can correct: deep-rollout states are under-penalised relative to their realised risk. **The limit, in the same breath:** this bounds what the quantity reports across depth, not what the distortion costs a trained policy, and no policy is trained here (§11).

The larger sample changes one thing materially, and it is a correction to our own earlier reading. At $n_{\text{independent}} = 4$ the epistemic ordering looked like chance at short horizon — 23 of 45 dimensions at $h=1$ — and we had described it as a long-horizon effect. At $n_{\text{independent}} = 20$ it is 44 of 45 at $h=1$, with mean $r = +0.662$, the *strongest* mean correlation of any horizon. The in-sample permutation test says the same (§6.6). The short-horizon "chance" result was an artifact of four trajectories, not a property of the model, and we record it as such rather than keeping the more interesting-sounding horizon story.

Two pre-registered checks on how these numbers are read, both committed before they were computed. M-62 asks whether any verdict depends on resampling 400-step trajectories rather than whole episodes, which two trajectories share. It returns **NO MOVE**: in the one arena with power at that level — all ten episodes, $n = 20$ falling to 10 — the pooled correlation’s interval widens by 27% and the double-demeaned one by a factor of 0.98, and neither crosses zero. The other 4 cells it names are out-of-sample, where an episode bootstrap has $n = 2$ and three distinct resamples; the rule said so in advance rather than discovering it, and they are reported as uninformative rather than as intervals.

M-63 asks whether the one-step failure is a few bad channels or all of them, since a pooled coverage is the unweighted mean of 45 per-dimension ones. It returns **UNIFORM**: the interquartile range across dimensions is 10.0 points against a 15-point threshold committed in advance, and the five worst dimensions carry well under the half of the shortfall that would have made it concentrated. **No channel is exempt**, which is what §6.3’s mechanism predicts: an objective whose optimum is $\sigma = 0$ has no reason to spare any dimension. The reading is coarse by construction — at $h = 1$ on 20 trajectories a per-dimension coverage moves in 5-point steps, and the rule fixed that limit before the run.

The released checkpoint is no longer the only ensemble measured. Three Arm A arms at ensemble size 5 (§6.7, 3 seeds, out-of-sample, $n_{\text{independent}} = 4$) give, averaged over seeds:

h	epistemic err/ σ [95% CI]	$\pm 1\sigma$ [95% CI]	$\pm 2\sigma$	dims $r > 0$
1	$2.1\times$ [1.7, 2.5]	35.93% [29.63, 42.22]	60.37%	35.0/45
8	$6.3\times$ [4.4, 7.5]	14.10% [11.02, 17.18]	26.62%	37.3/45
32	$8.0\times$ [5.9, 9.8]	10.79% [8.53, 13.71]	20.50%	34.3/45
100	$10.5\times$ [9.0, 11.5]	8.19% [6.80, 9.59]	16.11%	44.0/45
128	$11.0\times$ [9.5, 11.9]	7.79% [6.81, 8.77]	15.34%	44.0/45
368	$13.0\times$ [11.9, 13.9]	6.30% [5.84, 6.77]	12.51%	40.7/45

Our arms are **better calibrated than the released checkpoint and fail the same way**: $10.5\times$ over-confident at $h = 100$ against its $33.4\times$, with 8.19% coverage where a calibrated Gaussian gives 68.27%. §6.4 establishes that the two are the same architecture in the respect that matters here, so this is a comparison of like with like. Being an order of magnitude closer to calibrated is not being calibrated.

The last column of the table above gives permutation P-values over whole trajectories, not binomial ones, computed on the same 20 trajectories as the counts beside them; §6.6 explains why a binomial null is inadmissible here and how far it was wrong. **$h = 100$ carries the abstract, so it is tested rather than left blank**: an earlier draft ran the permutation test on the pre-revision five-horizon grid and printed "—" in the one row the headline rests on. These are six tests on one family and none survives Holm–Bonferroni across the arena’s 30 cells — the smallest is faithful (mse) $h=368$ at 0.0037 against a threshold of 0.001667. Read the column as a consistency check on direction, not as six independent findings.

The scalar penalty as actually applied — `means.std(0).sum(-1)` at `envs/base.py:166` — correlates **+0.605** with total absolute error over the rollout, 95% CI $[+0.545, +0.694]$ from a bootstrap over whole trajectories, `n_independent = 20` (7,360 pooled trajectory-step points). An earlier draft quoted this correlation with neither an interval nor an n . The interval resamples whole trajectories, not trajectory-step pairs, which would narrow it by about the square root of the rollout length.

6.3 Why the aleatoric head collapses: the optimum is $\sigma = 0$

This subsection explains the aleatoric column and only that column. Ensemble disagreement is not shaped by the mechanism below, and why *it* is miscalibrated is not established here.

It also supplies the alternative explanation promised in §6.1. The follow-up reads the low aleatoric value as reflecting "small stochasticity in the environment". The observation is correct and the reading is not: σ is low because $\sigma = 0$ is the optimum of the loss that trains it, and it would be low on any dataset, stochastic or not.

The state loss is squared error on a *sample* drawn from the predicted Gaussian, not a likelihood:

$$\mathcal{L} = \mathbb{E}[(\mu + \sigma\varepsilon - y)^2] = (\mu - y)^2 + \sigma^2$$

which is minimised at $\sigma = 0$ for any μ . There is no $\log\sigma$ term to oppose it. The bound term that appears to oppose it does not, because `max_logstd` is not an independent parameter — it is constructed as `min_logstd + exp(log_delta_logstd)`, so

$$\overline{\log \sigma_{\max}} - \overline{\log \sigma_{\min}} = \overline{\exp(\log \Delta_{\log \sigma})}$$

and `min_logstd` cancels algebraically, taking no gradient from that term. The floor the interval closes onto therefore freezes while the interval closes: a one-way ratchet.

The derivation above covers two terms, and the objective has 7. Its completeness rests on the other 5 being inert with respect to σ , and a reader should not have to take that on trust. Each term is therefore computed alone on one real batch and back-propagated alone, and the gradient reaching the $\log\sigma$ tower, `state_log_delta_logstd` and `state_min_logstd` is recorded. A term that cannot move σ produces exactly zero on all three.

loss term	live?	weight	where the reference computes it	$\partial/\partial \log\sigma$ tower	$\partial/\partial \log_delta_logstd$	$\partial/\partial \min_logstd$
<code>state</code>	live	1.00	<code>system_dynamics.py:270-289</code>	0.000325	0.0509	0.0703
<code>sequence</code>	dead	1.00	<code>system_dynamics.py:274-277</code>	0	0	0
<code>bound</code>	live	1.00	<code>system_dynamics.py:301-302</code>	0	0.2	0
<code>k1</code>	dead	0.10	<code>system_dynamics.py:223</code>	0	0	0
<code>extension</code>	dead	1.00	<code>system_dynamics.py:233-268</code>	0	0	0
<code>contact</code>	live	1.00	<code>system_dynamics.py:233-268</code>	0	0	0
<code>termination</code>	live	1.00	<code>system_dynamics.py:233-268</code>	0	0	0

4 of the 7 configured terms are live at all under the released configuration — `sequence_loss` is dead code, guarded by a `prediction_type` the reference sets to "single" on both paths, and `k1` and `extension` are zero because their dimensions are. Of the 7, exactly 2 reach σ : `state` and `bound`. The remaining 5 produce a gradient of exactly zero, which is a stronger statement than reading the code and concluding they do not matter (`results/e4_sigma_gradients.json`).

The derivation says this would happen on any dataset. That is testable, and until now it was only asserted — which matters, because it is the sentence that answers the follow-up’s own explanation. The follow-up attributes its low aleatoric term to "small stochasticity in the environment", and on the released CSV that reading and ours are observationally identical: the data may simply be nearly deterministic. Under a rule committed before the runs (M-50) we therefore built data where it is not.

Synthetic data whose true noise level is **known** and varies by a factor of 25 across the input range, with a non-constant true mean; the **same** bounded log- σ head as the released model — `MLPStateHead` unmodified, including the double-softplus clamp, the learnable `state_min_logstd` and `state_log_delta_logstd`, and the bound loss at its configured weight — trained under each objective in turn on 4,000 points for 12,000 iterations at 3 seeds. Nothing else differs between the arms.

Two ways this is not the released setting, stated because "unmodified" is a claim about the class and not about the instantiation. The head is built here over a **one-dimensional** state with no recurrent trunk in front of it, where the released one predicts 45 dimensions from a GRU. The trunk’s absence is deliberate — the question is about the head’s objective, and interposing a GRU adds a confound rather than removing one. The dimensionality has a consequence worth naming: the state loss sums over state dimensions, so at one dimension it is roughly $45\times$ smaller relative to the bound term than in the released path, and the bound term is the other term §6.3’s derivation is about. That makes this arm’s conditions *more* favourable to σ surviving, not less — the collapse happens anyway.

objective	median $\hat{\sigma}$ / σ_{true}	$\hat{\sigma}$ spread across the input range	slope of $\log \hat{\sigma}$ on $\log \sigma_{\text{true}}$
mse — the implemented branch	0.0460	1.002–1.003 $\times$	+0.000493
gaussian_nll — the authors’ unused branch	0.9779	1.02–3.67 $\times$	+0.1330

Ratios and slopes are means over 3 seeds; spreads are the range across them, because the mean of a spread hides which seeds recovered.

Under the implemented objective σ sits $22\times$ below the true noise and does not track it at all — a spread of $1.003\times$ where the truth spans $25\times$, and a slope below the 0.00309 the design can detect. Under the authors’ own branch, same data and same head, σ recovers the true level to a median ratio of 0.9779 and every one of the 3 seeds M-50 was discharged over clears the slope threshold. **Twenty seeds show that clearance is not general:** 11 of 20 clear it, so the all-seeds criterion would not have held at that sample. M-50’s verdict stands as returned over its own 3 and is not re-opened by more seeds (§8); what twenty establish is that the hedge below was necessary rather than cautious. **The recovering arm is seed-variable and M-50 said so before the runs:** its slopes span 0.0078–0.3537, a factor of 45, and two of 3 seeds recover a σ spread of only $1.02\times$ against the truth’s $25\times$. So what this experiment establishes is the **contrast** — one objective tracks the noise at all and the other does not — and not the magnitude of the recovery, which this training budget does not pin down. **OBJECTIVE-DRIVEN**, which is the verdict M-50 names for that pattern.

The statistic is the slope and not the correlation, and the reason is worth one sentence: a correlation is scale-free, so a $\hat{\sigma}$ that is essentially constant still returns a large one off its own numerical noise. Under a permutation null — the same data with the input-to-noise pairing destroyed — a head whose σ spanned $1.0004\times$ returned correlations as large as ± 0.24 , while its slope was $2e-05$. The detection threshold is set at the slope corresponding to a $1.01\times$ spread rather than at that noise floor, and the measured false-positive rate at zero signal is 0%.

What this does that the derivation alone could not. It removes the competing explanation rather than arguing against it. The stochasticity here is large, known, and input-dependent, and the collapse happens anyway. The design’s limit is stated in M-50 and holds: the dilution ladder detects the signal at

full strength and at no dilution below it, so this establishes that σ does not track the noise **at all**, not the magnitude of how badly.

We predicted the collapse from this algebra before training, then observed it. Across all 26 runs the collapse is linear in iteration count and its rate is nearly identical (Figure 3a). Rates are fitted on 20 of those runs: the 6 10,000-iteration runs are excluded from the rate statistics because they continue seeds already counted at 2,500 and would double-weight them. Figure 3(a) shows all 31 runs; Figure 3(b) plots only the 20 the rate is fitted on, so the scatter and the quoted statistic describe the same set.

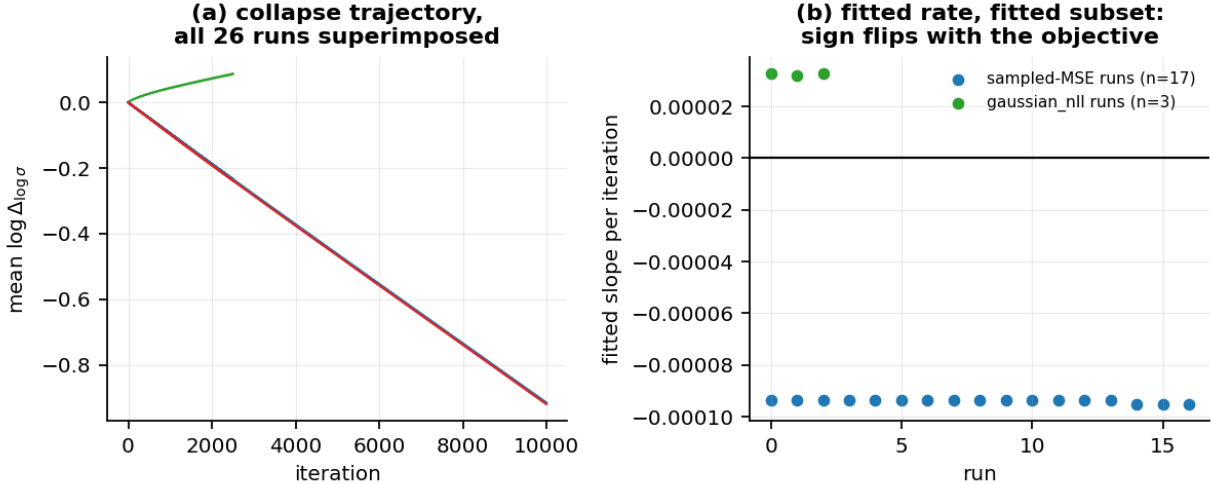


Figure 4: The variance collapse is objective-driven. (a) mean log $\Delta_{\log \sigma}$ against training iteration for every run. (b) the fitted per-iteration slope for each run, grouped by objective: negative and tightly clustered under sampled MSE, positive under `gaussian_nll`. The sign flip is the evidence that the objective, not the optimiser or the data, produces it.

The 31 runs, so a reader can count them:

arm	iterations	ensemble	objective	dataset	seeds	seed ids
Arm A	2,500	1	gaussian_nll	clean	3	0, 1, 2
Arm A	2,500	1	mse	clean	10	0, 0, 1, 1, 2, 2, 3, 3, 4, 4
Arm A	2,500	1	mse	contaminated	3	0, 1, 2
Arm A	2,500	1	mse	duplicated	3	0, 1, 2
Arm A	2,500	5	mse	clean	3	0, 1, 2
Arm A	10,000	1	mse	clean	3	0, 1, 2
Arm B	2,500	1	mse	clean	3	0, 1, 2
Arm B	10,000	1	mse	clean	3	0, 1, 2

Two different things are being explained here, and §6.6 separates them. *Magnitude collapse is objective-driven.* It occurs in all 17 sampled-MSE runs at a rate of $-9.3857\text{e-}05$ per iteration with a standard deviation of $6.0\text{e-}07$ — **including the teacher-forced arm**, which shares the objective — and reverses to $+3.2332\text{e-}05$ in the 3 runs that change it. *Input-independence is not.* That varies by a factor of 15.6 between two arms trained under the same objective, so the objective cannot be what produces it.

Under the corrected objective the sign flips (Figure 3b) — which is the strongest evidence that the mechanism is the objective and not the optimiser, the data or the architecture.

6.4 Why the epistemic term may be miscalibrated: the members are not independent models

§6.3 explains the aleatoric column and says of the epistemic one that the mechanism is not established. This subsection supplies a candidate, structurally symmetric to §6.3's, from source and from the checkpoint's own tensors. Nothing here is trained and nothing is inferred from a measurement.

The effect is known and we are not claiming it. That heads sharing a trunk under-report disagreement relative to independently initialised networks is established: Lee, Purushwalkam, Cogswell, Crandall and Batra (arXiv:1511.06314) treat ensemble diversity as something to be engineered rather than assumed; Fort, Hu and Lakshminarayanan (arXiv:1912.02757) show that what independent initialisation buys is decorrelation which subspace methods do not match; and BatchEnsemble (Wen, Tran and Ba, ICLR 2020) and MIMO (Havasi et al., ICLR 2021) share deliberately and state what they trade away for it. **What is ours is finding it in a released robotics checkpoint that its authors deployed on hardware, with the sharing quantified and the cost measured** — 89.15% of each member, and $2.03\times$ on the overconfidence factor (§6.10). The problem is not sharing. It is sharing and then reading the spread as though the members were independent.

The released five-member ensemble is not five models. `system_dynamics.py:34` builds `one_state_base`. `system_dynamics.py:35-41` replicates the `heads_ensemble_size` times, and only the heads. In the forward pass `system_dynamics.py:87` evaluates the trunk **once** and `:90` hands the identical feature vector to every head; `system_dynamics.py:126` then computes the epistemic term as the standard deviation across those heads.

The parameter counts make the scale of the sharing concrete. The state pathway is a 636,672-parameter two-layer GRU trunk plus five heads of 77,492 parameters each. Per member, 636,672 of 714,164 parameters — **89.15%** — are numerically identical to every other member's. Only 10.85% differ. Across the whole released object, the two shared trunks are 63.81% of 1,995,569 parameters.

The sharing is stronger than the parameter count suggests, and this is the part that matters. The trunk owns a *single* recurrent hidden state (`rnn.py:40`), and an autoregressive rollout feeds the ensemble **mean** back into it (`system_dynamics.py:115`; `src/rwm_model.py:223` in our reimplementation). So the five members do not roll out independently at all. There is exactly one hidden-state trajectory between them, and disagreement at step t is the spread of five two-layer MLPs read off a single 256-dimensional vector, at whatever point that one trajectory has reached.

That is the argument. **Members which share a feature extractor have correlated errors by construction, and their spread cannot express uncertainty the shared trunk does not already carry.** Where a deep ensemble in the sense of Lakshminarayanan et al. varies initialisation *and* data ordering across whole models, here only the output heads differ — so the quantity the method penalises rewards with is a lower bound on epistemic uncertainty by construction, not by accident. It is the same shape of finding as §6.3: not a training failure, a structural one.

This applies to our own arms identically, which is why §6.2's comparison is fair. Our ensemble-5 arms build one trunk the same way (`src/rwm_model.py:164-167`), evaluate it once (`:182`), hand the same vector to every head (`:185`) and compute the same spread (`:200`). Their tensor names and parameter counts match the released checkpoint exactly — 636,672 shared, 77,492 per head, 1,995,569 in total, on all 3 arms checked. So §6.2's "our arms fail the same way at $10.5\times$ at $h = 100$ " compares two instances of one architecture, not two architectures. 21 source citations support the paragraphs above and each is read back from the pinned upstream and checked on every build (`results/v1_ensemble_topology.json`).

What this is and is not. It is a *candidate* mechanism, established structurally. It is not yet a demonstration that trunk-sharing is *the* explanation for the miscalibration in §6.2 — architecture could be a minor contributor to a failure dominated by something else. Establishing that needs a comparison against an ensemble which shares nothing, which is what M-44 pre-registers and what §6.10 reports. We keep the topology and the mechanism separate on purpose: the topology is a fact about the released artifact, and the mechanism is a hypothesis about that fact.

6.5 The correction fails differently rather than succeeding

The reference contains an unused `gaussian_nll` branch. Running it reverses the collapse and improves the magnitude from $52.2\times$ to $10.9\times$ overconfident. It does not produce a usable estimate, and it destroys something the faithful arm had: the σ -versus-error ordering falls from 39/45 dimensions positively correlated to 21/45, which is chance. Under the trajectory permutation test of §6.6, at $h = 368$, those counts give $P = 0.0417$ and 1.0000 out of sample, 0.0240 and 0.8746 in sample. The faithful arm’s ordering is the one result in this family that points the same way in both arenas; it is also the weakest effect of the three, and it does not survive multiplicity correction either.

6.6 The failure is one of magnitude; the ordering is weaker than it looks

Measuring the teacher-forced arm — which we had trained for §5, and which our own first three calibration tables omitted — sharpens the finding:

model	σ variation across inputs (CoV)	dims with $r(\sigma, \text{error}) > 0$ at $h=368$, out-of-sample	perm P at $h=368$, out-of-sample	perm P at $h=368$, in-sample
faithful Arm A	0.0076	39/45	0.0417	0.0240
corrected Arm A	0.0059	21/45	1.0000	0.8746
teacher-forced Arm B	0.1188	45/45	0.2609	0.5565
released checkpoint	0.0177	20/45	0.6957	0.9753

The CoV column is the aleatoric σ in every row, which is the only σ the ensemble-size-1 arms have. Our ensemble-5 arms have both: their aleatoric CoV is comparable to the other arms’, and their *epistemic* term is far more input-dependent than any aleatoric head here, at $0.379\text{--}0.395$ against the released checkpoint’s 0.0177 (§6.7). **The count column is the out-of-sample arena** ($n_{\text{independent}} = 4$), so that all four models are compared on trajectories none of our own arms was trained on. It is not the only arena, and for the released checkpoint’s aleatoric head it is not the most informative one: at $h = 368$ and $n_{\text{independent}} = 20$ over all ten episodes that head is $0/45$ — negatively correlated with error on *every* dimension — against $20/45$ here. §12 quotes the larger arena and says so.

Arm B’s σ is $15.6\times$ more input-dependent than the faithful arm’s, and it has the largest mean correlation of the four ($r = 0.257$). It is still $315\times$ overconfident.

The P column above is not a binomial one, and an earlier draft of this paper was wrong to make it one. Converting a count of positive per-dimension correlations to a P-value against a fair-coin null assumes the 45 state dimensions are independent trials. They are not. Position, velocity and torque for the same joint are physically coupled, and base linear and angular velocity are coupled through the gait. More importantly, error grows with rollout depth in every trajectory, so *any* σ that also grows with depth correlates with *any* trajectory’s error — including one it was never paired with.

We therefore permute whole trajectories. The null pairs each trajectory’s σ with a different trajectory’s realised error, which leaves both marginal distributions and the entire cross-dimension dependence structure intact and destroys only the association under test. The correction is large, and it is largest exactly where we leaned hardest. The worst-affected cell is teacher-forced armB at $h=368$, in the in-sample arena. At $h = 368$ it moves from $5.68e-14$ to 0.5565 — a factor of about 10^{13} — because under a null that preserves the dependence, a random re-pairing already yields 43.8 of 45 dimensions positive on average. Observing 45 of 45 against that null is close to unremarkable. A fair coin, by contrast, centres the count at 22.5 of 45 ; the dependence-preserving null centres it between 5.1 and 43.8 depending on model, horizon and arena.

So σ *collapsing in magnitude* is objective-driven, and σ *becoming input-independent* is not. The teacher-forced arm collapses in magnitude exactly like the autoregressive ones — same objective, same rate — while retaining $15.6\times$ more variation across inputs. Input-independence is a property of the autoregressive arms and the released checkpoint, and input-dependence and correct ranking are both achievable without the interval becoming meaningful.

One candidate mechanism, stated as a hypothesis and not a result: autoregressive feedback narrows the input distribution toward the model’s own manifold, leaving a heteroscedastic head less variation to key on. We have not tested it.

The same pattern holds for the quantity the method uses, and this is where the correction bites hardest. At $h=128$ and $h=368$ the epistemic term correlates positively with realised error on **45 of 45** dimensions, matching the best aleatoric head here on the sign count, while being $39.7\times$ overconfident at $h = 368$. **All figures in this paragraph are the held-out arena ($n_{\text{independent}} = 4$),** so that the epistemic term and the four aleatoric heads are compared on identical trajectories; §6.2 quotes $34.4\times$ for the same ratio at $h = 368$ and $n_{\text{independent}} = 20$. The figure the abstract and §12 use is neither of those: it is $33.4\times$ at $h = 100$, on the same 20 trajectories. It does not beat Arm B’s head on strength either: its mean correlation at $h=368$ is $+0.151$ against 0.257 . The two quantities rank comparably; neither is close to an interval. Under the permutation null that count gives $P = 0.0435$ out of sample and 0.0775 in sample, against $5.68e-14$ from the independent-trials test we should not have used. It still fails the horizon test the same way: σ grows $1.59\times$ from $h=1$ to $h=368$ while error grows $13.33\times$.

The horizon story we first told was backwards, and the larger arenas agree with each other against the smallest. At $n_{\text{independent}} = 4$ out of sample, the epistemic ordering looked strongest at long horizon (0.0417 at $h=128$, 0.0435 at $h=368$) and unremarkable at short (0.4348 at $h=1$). Both larger arenas invert that. In sample ($n_{\text{independent}} = 16$): 0.0052 at $h=1$, 0.0070 at $h=8$, against 0.3794 at $h=128$. Over all ten episodes ($n_{\text{independent}} = 20$): 0.0056 , 0.0069 and 0.3762 , with $h = 100$ at 0.2769 sitting between $h=32$ ’s 0.0344 and $h=128$ ’s 0.3762 — the horizon added by this revision falls where the existing reading says it should, which is worth stating because it was not free to. Two independent arenas at four and five times the sample say the effect is strongest at *short* horizon.

The null means explain why, and the explanation is the same one that motivates §6.7. At long horizon the shared forecast-depth trend lifts the null to 41.6 of 45 at $h = 128$, so a count of 45 is close to what chance alone delivers; at $h = 1$ the null sits near 15.4 and the same count is genuinely surprising. The out-of-sample arena is not wrong so much as blind: at 4 trajectories its smallest attainable P-value is 0.04167 , so at this unit it cannot distinguish a strong effect from a marginal one at any horizon. **That blindness is a property of the 400-step unit, not of the arena,** and the distinction now matters: a 33-row unit gives 60 independent units on the same two episodes, and 14 at $h = 100$ (M-64). **We did not recompute this permutation family at the shorter unit,** so what the short units change here is the scope of the design claim rather than any verdict in it: the arena is demonstrably underpowered at $h = 368$, which needs the full 400 rows, and is no longer demonstrably underpowered at $h = 128$ and below. Running it there is one pass over stored rollouts and we did not do it. We report the small arena’s numbers alongside because it is the only arena that is out-of-sample for our own arms, not because it is the better measurement.

Nothing here survives multiplicity correction, in any of the three arenas. Holm–Bonferroni over each arena’s 30 model $\times$ horizon cells at $\alpha = 0.05$ rejects 0 out of sample, 0 in sample and 0 over all ten episodes. Out of sample that is a property of the design rather than of the models: with 4 independent trajectories the smallest attainable P-value is 0.04167 , which already exceeds the smallest Holm threshold 0.001667 , so no effect of any size could have been rejected there. In sample the miss is real — the smallest P in the family is teacher-forced armB $h=128$ at 0.0027 against a threshold of 0.001667 .

So the honest form of this section’s claim is narrower than the one we first wrote. **The magnitude failure is established and large; the ordering is directionally consistent across every model and horizon we measured, and is not established at conventional significance once the dependence between dimensions is respected.**

The failure is specifically magnitude calibration, in both components.

6.7 Ensemble disagreement beats the trivial baseline

The follow-up justifies ensemble disagreement as a trust metric on the grounds that it "closely follows the trend of the prediction error". Section 6.6 shows the per-dimension version of that claim is weaker than it looks. This section asks a different and, for a practitioner, more important question: **does disagreement beat something free?**

Error in an autoregressive rollout grows with depth. So the trivial competitor to any trust metric is the forecast step index — a counter. It needs no ensemble, no second forward pass and no model. If a counter

ranks error as well as disagreement does, the ensemble is not earning its cost. Neither paper runs this comparison, so we do.

All three correlations below are on the scalar quantity the method actually applies — `means.std(0).sum(-1)` at `envs/base.py:166` — against total absolute error, over `n_independent = 20` trajectories, with 95% intervals from a bootstrap over whole trajectories.

h	r(step index, error)	r(disagreement, error)	partial r(disagreement, error · index)	paired difference, disagreement − index
1	—	+0.994 [+0.918, +0.999]	—	—
8	+0.181 [+0.108, +0.349]	+0.738 [+0.537, +0.807]	+0.757 [+0.540, +0.822]	+0.556 [+0.241, +0.679]
32	+0.174 [+0.131, +0.336]	+0.735 [+0.577, +0.823]	+0.756 [+0.604, +0.841]	+0.561 [+0.298, +0.674]
100	+0.490 [+0.394, +0.604]	+0.628 [+0.574, +0.839]	+0.584 [+0.516, +0.815]	+0.138 [+0.034, +0.362]
128	+0.526 [+0.421, +0.643]	+0.671 [+0.604, +0.846]	+0.617 [+0.534, +0.812]	+0.145 [+0.011, +0.324]
368	+0.269 [+0.106, +0.431]	+0.605 [+0.545, +0.694]	+0.596 [+0.526, +0.687]	+0.337 [+0.147, +0.544]

(*h=1 has a single forecast step, so the index is constant and its correlation — and therefore the partial and the difference — undefined. The epistemic correlation is not, and it is the largest anywhere in this work: at one step, ensemble disagreement is very nearly a perfect ranking of realised error.*)

Disagreement wins at every horizon tested. Over the full $h = 368$ rollout the counter reaches +0.269 against disagreement’s +0.605, and the index leads in 0 of 5 horizons.

The last column answers the first question — does disagreement beat the counter — and it is not the test a reader might expect. Comparing the two marginal intervals for overlap is the wrong comparison here: both correlations are measured on the *same* trajectories, so their sampling errors move together and the marginal intervals are needlessly conservative. The paired difference — resampling whole trajectories and recomputing *both* correlations inside each draw — is the appropriate test and the more powerful one. It excludes zero at **5 of 5** horizons.

The distinction matters at 2 of the 5 horizons. At $h=100$ and $h=128$ the marginal intervals *do* overlap, and an earlier draft of this paper wrongly asserted that they never do. $h=128$ is where the counter is strongest (+0.526) and the margin narrowest: the paired difference there is +0.145 [+0.011, +0.324], which excludes zero, but only just — +0.011 is the smallest lower bound in the table and we would not rest anything on that horizon alone — at the 400-step unit. At the shorter unit M-64 builds, 60 non-overlapping units over the same ten episodes give [+0.148, +0.351] for the same paired difference, so the horizon that carried the least weight carries more than it did. At $h = 100$ the paired difference is +0.138 [+0.034, +0.362], which also excludes zero.

The third column answers a different question: is disagreement merely re-encoding the clock? Partialling the step index out of both variables *lowers* disagreement’s correlation by 0.010, from +0.605 to +0.596. Almost none of what disagreement knows is explained by knowing how deep into the rollout you are. It is carrying real information about *this* rollout, not a re-encoding of the clock.

A linear control is not much of a control, so we tested four harder ones. Error does not grow linearly with rollout depth, and a control that under-fits the index leaves index-driven variance in the residual and flatters disagreement. Partialling out $\log(1 + \text{index})$ gives +0.589 [+0.512, +0.688]; a cubic in the index gives +0.582 [+0.495, +0.676]; a rank partial correlation, which removes *any* monotone dependence on depth rather than an assumed functional form, gives +0.906 [+0.856, +0.921].

The strongest of the four needs no model of the index-error relationship at all. Computing the correlation **within each forecast step** — across trajectories, with depth held exactly constant, so the index cannot contribute by construction — and averaging over steps gives **+0.739 [+0.711, +0.852]**, positive at 368 of 368 forecast steps with a median of +0.737.[$\hat{\text{stepcount}}$] The weakest figure across all 5 controls is +0.582. Disagreement is not re-encoding the clock: at a fixed depth it still knows which rollouts are going wrong.

[$\hat{\text{stepcount}}$]: Adjacent forecast steps on the same 20 trajectories are heavily dependent — structurally the same problem §6.6 spends a page correcting for the 45 coupled state dimensions. The count is descriptive; the interval [+0.711, +0.852] is the statistic, and no P-value attaches to 368/368.

All five of those controls remove forecast depth. None removes trajectory difficulty, and the sixth does. This is the one control in this section that was pre-registered before the statistic was computed (M-45, §8), and adding it changed how we read the five above.

Per-episode difficulty in this dataset spans 0.562 to 1.591 and is uncorrelated with commanded speed, so the units being correlated differ a great deal in *level*. If harder trajectories simply have both larger error and larger disagreement, the pooled correlation would look exactly as it does with disagreement carrying no within-rollout information whatever. Two things suggested that was worth checking: the 20-point $h = 1$ figure of +0.994, which is not the shape of a genuine per-step signal, and the within-step control coming out *above* the pooled figure, which is the signature of a between-unit effect.

Splitting the pooled correlation settles which effect carries it:

what is removed	correlation	95% CI	what it answers
nothing (pooled)	+0.605	[+0.545, +0.695]	—
everything within a trajectory — the 20 trajectory means alone	+0.878	[+0.817, +0.956]	do harder rollouts disagree more?
forecast-step means only	+0.589	—	the existing within-step control, pooled
trajectory means only	+0.470	[+0.389, +0.604]	—
both, additively (r_dd)	+0.419	[+0.318, +0.576]	at a given depth, in a given rollout, does disagreement know?

The between-trajectory correlation is **+0.878**, and the two components contribute 51.7% and 48.3% of the pooled covariance. So a large part of what §6.7 has been reporting is a between-rollout effect. **That reinterprets the fifth control rather than merely adding to it.** The within-step statistic correlates *across trajectories* at each fixed step and averages — which removes depth completely and removes trajectory difficulty not at all. It is a mean of 368 between-trajectory correlations, which is why it reads +0.739 against the pooled +0.605 rather than below it. It was described in an earlier draft as "the decisive one"; it is not, and we withdraw that description.

The decisive statistic is the double-demeaned one, and it survives. Removing the trajectory mean *and* the step mean from both variables and correlating the residuals gives **+0.419 [+0.318, +0.576]**, on $n_{\text{independent}} = 20$ with a cluster bootstrap over whole trajectories. M-45's threshold, fixed before the statistic was computed, was that this interval exclude zero. The rule's minimum detectable effect at this sample size is an absolute r_dd of 0.183 or more, estimated by a dilution study which put the detection threshold between a true effect of 0.086 (not detected) and 0.172 (detected). The observed effect is comfortably above it. **M-45 returns SUPPORTED:** with both the rollout and the depth held constant, disagreement still tracks error. It is not merely reporting which episode is hard.

Per horizon, on the same 20 trajectories and the same cluster bootstrap:

h	r_dd, double-demeaned	95% CI	excludes zero
1	—	—	—
8	+0.306	[-0.387, +0.713]	no
32	+0.186	[-0.032, +0.636]	no
100	+0.385	[+0.283, +0.701]	yes
128	+0.453	[+0.359, +0.713]	yes
368	+0.419	[+0.315, +0.573]	yes

($h = 1$ has one forecast step per trajectory, so there is nothing within a rollout to demean against and r_dd is undefined rather than zero.)

Two qualifications a reader should carry away with that. The within-rollout effect is **materially smaller than the pooled figure** — +0.419 against +0.605 — so a practitioner should expect disagreement to separate *rollouts* better than it separates *moments within a rollout*. And it is **not established at short horizon**: r_dd's interval excludes zero at $h=100$, $h=128$, $h=368$ and spans zero at $h=8$ and $h=32$, where too few steps exist to demean against. That inverts the shape one might expect and we report it as measured. At $h = 100$ it is established, at +0.385 [+0.283, +0.701].

The $h = 1$ figure survives the same test, but it is not what it looked like. At $h = 1$ the panel has one column, so +0.994 is a correlation over 20 *trajectory-level* points and nothing within a rollout is being

tested at all. We checked whether trajectory difficulty manufactures it: disagreement correlates $+0.006$ with commanded speed and $+0.040$ with per-episode difficulty, and partialling both out of the disagreement–error correlation leaves $+0.995$ — it does not move. So the figure is real and is not a difficulty artifact. It is nevertheless a statement about **ranking whole rollouts at one step ahead**, on 20 points, and §9 now says that rather than calling it a ranking of realised error without qualification.

Does it hold on a model we trained? Everything above is measured on the released checkpoint, because our main arms run at ensemble size 1 where the epistemic term is identically zero. We therefore trained three Arm A arms at **ensemble size 5**, identical in every other setting, under a rule committed to git before the runs existed (§8, M-43). The rule asked for two things: that disagreement lead the index at every horizon, and that the paired difference exclude zero at a majority of them.

It returns DOES NOT GENERALISE. The first condition passes completely — disagreement leads the index in **12 of 12** seed-horizon cells, every paired estimate positive, $+0.204$ to $+0.545$. The second fails: the paired difference excludes zero at 1 of 4 horizons, not a majority. We report the verdict the rule returns and do not rewrite the rule.

And we do not rewrite its denominator either, which is the less obvious half of the same discipline. M-43 was committed over 4 horizons, before the data. Adding $h = 100$ to the evaluation grid after the fact would change what "a majority of horizons" means in a rule already discharged — a way of moving a threshold that looks like reporting rather than like moving a threshold. The verdict above is over M-43's own

1. The released checkpoint's table in §6.7 does follow the six-horizon grid, because no pre-registration is stated over it; the two counts are deliberately different numbers and the build keeps them in separate keys for that reason.

What separates the two conditions is sample size, and we measured that rather than asserting it. Our own arms can only be scored out-of-sample on the held-out pair, $n_independent = 4$, where §6.7's own finding used 20. Subsampling four trajectories at a time from a twenty-trajectory pool, the rule's criterion fires on 75% of draws on average and on only 24% at $h=8$. That estimate is an **upper bound**, because the pool it subsamples is in-sample for these arms, where the effect is $+0.425$ to $+0.790$ against $+0.204$ to $+0.545$ on the held-out pair. So the rule was under-powered at the sample size it faced, decisively at one horizon — and we do not claim it could not have passed, only that it was committed without anyone checking what it could detect. That is a failure of ours, and it is the same one the ledger already records as M-24: a rule anchored without regard to the regime it would be applied in.

Reported as a companion and not as a discharge: on all ten episodes ($n_independent = 20$, **in-sample** for these arms, which trained on eight of them) the same measurement excludes zero at 4 of 4 horizons and would have satisfied both conditions. It cannot discharge M-43, which is stated over the out-of-sample arena, and we record it only so the comparison with the released checkpoint's 20 is like for like.

We ran the baseline test expecting it to go the other way. A counter matching disagreement would have been the more consequential result — it would make the trust metric close to vacuous, since a counter is free — and that is the outcome this test was set up to expose. We record the expectation as an expectation only: it was not committed to git before the data existed, so by this paper's own standard (§8) it is not a pre-registration, and it carries none of the weight one would. It did not go that way against the counter. It went that way against something else.

One adversary is one, so we added two more — and the ranking claim survives only one of them. A claim that beats exactly one competitor is a claim about that competitor. Under a rule committed before either was computed (M-51, corrected by M-52), we added two further baselines needing no ensemble and no second model: **step-size**, the magnitude of the model's own predicted state change $\|\mu_t - \mu_{t-1}\|$, which costs nothing because the rollout has already made those predictions; and **entry-res**, its one-step error at the step *before* the forecast window opens, which costs **one extra rollout in this harness** and nothing in deployment, where a model consumes the history to build its recurrent state anyway. M-51 called both free without distinguishing those, and M-52 records the correction.

baseline	r(baseline, error)	margin	partial r(disagreement given baseline)	beaten?
forecast step index (<i>the existing adversary</i>)	+0.2686	+0.3368	+0.5957	—
step-size — $\ \mu_t - \mu_{t-1}\ $	+0.4697	+0.1357	+0.5430	no
entry-res — one-step error before the window	+0.1471	+0.4582	+0.5948	yes

$r(\text{disagreement}, \text{error}) = +0.6053$ on the released checkpoint at $n_{\text{independent}} = 20$. Minimum detectable effect, estimated before either baseline existed: 0.2891 on a margin, 0.1131 on a partial.

The verdict is SURVIVES entry-res ONLY, and the reason is the row a reader should look at twice. The magnitude of the model’s own predicted state change ranks realised error at +0.4697, against +0.6053 for the five-member ensemble disagreement the method is built on. The margin between them, +0.1357, is **below** the 0.2891 this sample size can resolve — so at $n_{\text{independent}} = 20$ we cannot say the ensemble ranks better than a subtraction.

Disagreement does still carry information the subtraction does not. With **step-size** partialled out it retains +0.5430, far above the 0.1131 MDE for that test, and holding the forecast step exactly fixed it keeps +0.7392 against **step-size**’s +0.5196. It is not re-encoding predicted step size. What is not established is that it adds enough to be worth five models.

M-51 fixed this reading in advance rather than after the fact: the partial is the load-bearing test and the margin is corroboration, so a baseline passing the partial while failing the margin does not refute the ranking claim. **It does refute a framing.** "Disagreement ranks error well" is supported. "You need the ensemble to rank error" is not, and this paper said the second thing in an earlier draft.

On this axis the follow-up’s claim survives adversarial testing against a real baseline, and that is the strongest form of support this paper offers any claim of either original work — now with the qualification that a free baseline comes closer to it than the counter did. It coexists with §6.6 without contradiction: the *scalar* the method applies tracks error well, while the *per-dimension* sign counts we had leaned on carry far less evidence than an independent-trials test suggested. The quantity is a usable ranking signal and is still not an interval.

A note on the undefined cell. The within-step control holds the forecast step fixed and correlates across trajectories, so it annihilates any quantity that is constant across trajectories at a fixed step — which is the forecast index, and its within-step correlation is therefore undefined rather than zero. It does **not** annihilate a per-trajectory scalar like **entry-res**, which varies across exactly the axis the control varies over. M-51 said the opposite and M-54 records the correction.

6.8 One constant scalar does not fix it, but a per-horizon one does

Recalibrating a dynamics model’s uncertainty inside model-based RL is not new, and this section is an instance of it rather than a departure from it. Kuleshov, Fenner and Ermon (ICML 2018) give the post-hoc calibration map; Malik, Kuleshov, Song, Nemer, Seymour and Ermon (ICML 2019) apply that idea to model-based RL specifically and argue there that good uncertainties must be calibrated rather than merely well ranked — which is the distinction this paper draws throughout §6.2 and §6.7. **What is new here is the conditioning variable and one negative result.** The variable is the *forecast horizon*, and the negative result is that a single global multiplier **fails** where a per-horizon one works. That is not a detail: an open-loop rollout’s error accumulates with depth while its predicted σ does not (§6.9), so a horizon-blind recalibration cannot follow the thing it is trying to correct. This section is the measurement showing it does not.

If σ had the right shape and the wrong scale, a single multiplier would repair it, and the finding would be a units problem with a one-line remedy. We tested that. A scalar was fitted on **one** held-out episode and evaluated on the **other**, in both directions, so it is never fitted on its own test set.

Fitting at one step works at one step and nowhere else. On the epistemic term — the quantity the method uses — a scalar of 5.08–5.82 brings $h=1$ coverage to 63–74%, essentially calibrated against the 68.27% target, and leaves $h=368$ at 17–21%. On the aleatoric term a scalar of 593–611 gives 64–70% at $h=1$ and 11–15% at

$h=368$. Fitting over the whole rollout instead drives one-step coverage to 100% — an interval wide enough to be vacuous where the model is accurate — while still falling short at the far end.

The reason is §6.9’s mechanism: a constant multiplier cannot track an error that grows while σ does not. So "right shape, wrong scale" is the charitable reading of these tables, and for a *constant* scale it does not survive.

A per-horizon scalar does work, and this is the one concrete remedy in this paper. Fitting one multiplier per horizon on one held-out episode and evaluating on the other, in both directions, so no multiplier is ever scored on the episode that produced it. The two held-out episodes contribute 4 non-overlapping 400-step trajectories between them, so each direction fits on $n_independent = 2$ and is scored on the other 2:

quantity	held-out cells within 10 points of 68.27%, per-horizon c	same, constant c	range of fitted c
aleatoric	12 / 12	2 / 12	592.6 – 7782 (13.1 $\times$)
epistemic	12 / 12	2 / 12	5.082 – 47.33 (9.31 $\times$)

Every held-out cell lands within 10 points of the 68.27% target for both quantities. The largest deviation over all 24 held-out cells is aleatoric at $h=100$, fitted on episode 8 and scored on the other, at 77.48% — 9.21 points off target. The two largest deviations are both on the aleatoric term and both above target — 77.48% at $h=100$ and 76.55% at $h=128$ — so the fitted multiplier is mildly **conservative** at the long horizons rather than unstable in both directions. An earlier draft described them as opposite; they are not, and the **sign** family of checks now covers this sentence. The constant scalar manages 2 of 12, and those are the $h=1$ cells it was fitted at.

Three cautions a reader should apply. The per-horizon scalar has one free parameter per horizon against the constant one’s one, so it *must* fit better in sample — only the held-out column above is evidence, and that is the column reported. **And the held-out column is thinner than its count suggests:** the 12 cells are 6 horizons $\times$ two fold directions on the same 4 trajectories, and each multiplier is fitted on $n_independent = 2$ and scored on the other 2. They are not 12 independent successes and no P-value attaches to the count; it is reported so a reader can see how thin the evidence is, alongside a result we believe. And the correction is a calibration patch, not a fix: it leaves the model’s σ carrying no more information than before and simply rescales it by how far ahead you are looking. It is nevertheless enough to make the interval mean what it says, which is what a downstream user needs, and it costs one lookup table.

So the accurate form of this section is: **a constant scalar does not repair the interval; a per-horizon one does, and transfers across episodes.**

6.9 The structural excuse does not survive

One could argue that a model trained on an 8-step horizon cannot be expected to report calibrated uncertainty about step 368. It cannot report it about step 8 either. Inside the trained horizon, σ is flat while error grows (Figure 2):

model	σ growth, step 1 $\rightarrow$ 8	error growth, step 1 $\rightarrow$ 8
faithful Arm A	0.9241 $\times$	3.49 $\times$
corrected Arm A	1.0003 $\times$	3.41 $\times$
teacher-forced Arm B	1.0096 $\times$	6.11 $\times$
released checkpoint	1.0007 $\times$	1.79 $\times$

The faithful arm’s σ *declines* (0.9241 $\times$) while its error grows 3.49 $\times$. The coverage collapse in Figure 1(b) is therefore driven entirely by growing error against a fixed σ .

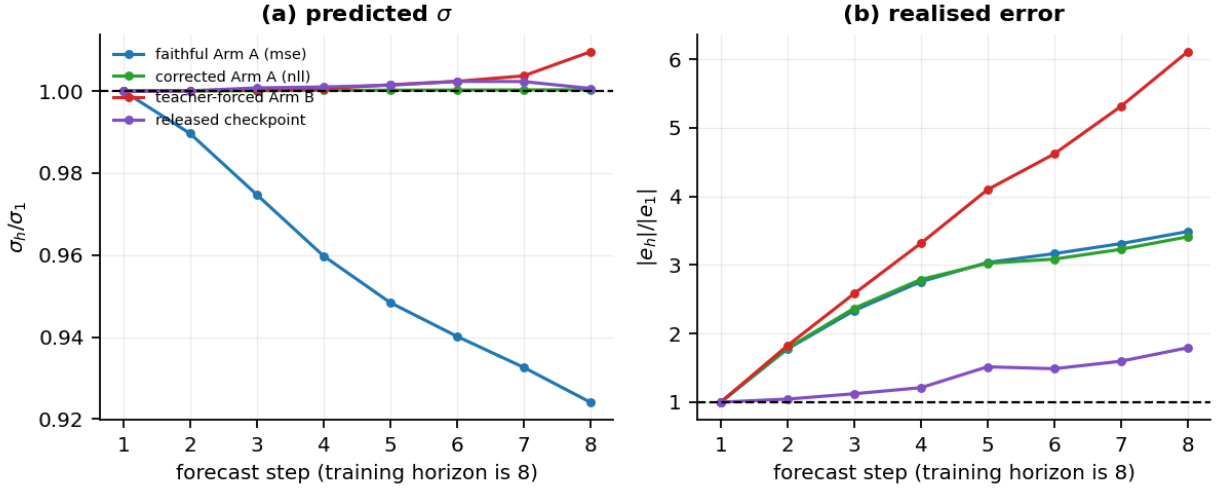


Figure 5: Why the coverage collapse is a horizon effect. Both panels are normalised to forecast step 1. (a) predicted σ barely moves, and for the faithful arm it declines. (b) realised error grows by an order of magnitude over the same steps. The gap between the panels is the collapse.

6.10 Testing the mechanism: an ensemble that shares nothing

§6.4 establishes the topology as a fact and the mechanism as a hypothesis. This subsection tests the hypothesis, under a rule (M-44) committed to git before any of the artifacts below existed, together with a power check estimating what that rule could detect at the sample size it would face.

The contrast, and why it is affordable. Training 5 genuinely independent models from scratch costs about 4.8 h of wall clock on two cores at the iteration count these runs use — 4.8 h against Appendix B’s 48.2 h for the whole project. Arm A at ensemble size 1 already existed at seeds 0, 1 and 2; we added two more at about 0.9 h each, 1.7 h in total, and scored the 5 together as an ensemble **at evaluation time**. No new training code and no new architecture — and the disagreement across 5 independently-initialised *full models* is exactly the contrast §6.4 asks for. The rollout protocol mirrors the shared-trunk one in every respect except the one under test: each member sees the same input state, each keeps **its own** recurrent hidden state, and the ensemble mean is fed back to all of them.

h	independent err/ σ	shared-trunk err/ σ	independent $\pm 1\sigma$	shared-trunk $\pm 1\sigma$
1	1.4×	2.1×	50.00%	35.93%
8	3.4×	6.3×	22.64%	14.10%
32	4.4×	8.0×	17.92%	10.79%
100	5.2×	10.5×	15.31%	8.19%
128	5.3×	11.0×	15.03%	7.79%
368	5.1×	13.0×	15.00%	6.30%

Same trajectories, same harness, $n_{\text{independent}} = 4$. The shared-trunk column is the mean over 3 seeds; the comparison below is paired against each of them separately.

M-44 returns MECHANISM SUPPORTED. All 6 of its 6 conditions hold, against every one of the 3 shared-trunk seeds. At $h = 100$ the independent ensemble’s overconfidence factor is $0.485\text{--}0.503\times$ the shared-trunk arms’ — a **2.03×** improvement against a pre-registered minimum detectable effect of $1.45\times$ — and its $\pm 1\sigma$ coverage is 6.69 to 7.33 points higher, a mean of +7.12 against an MDE of 2.26. Every paired interval excludes zero. **Members that share a feature extractor do produce a smaller spread, and the effect is large enough to matter.**

Where the improvement comes from, which is not all one thing. The overconfidence factor is error over σ , so it improves if σ grows *or* if error shrinks — and only the first is the trunk-sharing mechanism. Five independent models also denoise better than five heads on one trunk, which is an ordinary ensembling effect and not the thing under test. Splitting the improvement into its two multiplicative parts:

h	σ larger by	error smaller by	total	share from σ	share from accuracy
1	1.56×	0.97×	1.52×	106%	-6%
8	1.81×	1.02×	1.84×	97%	3%
32	1.68×	1.08×	1.83×	87%	13%
100	1.65×	1.23×	2.03×	71%	29%
128	1.62×	1.29×	2.09×	65%	35%
368	1.49×	1.70×	2.53×	43%	57%

*Shares are of the log improvement, so they add to 100%. A share above 100% at $h=1$ means the independent ensemble is very slightly the **worse** predictor there and the σ gain more than covers it.*

The reading, stated at the horizon the rule is stated over. σ is larger at every horizon — by 1.49×

 at its weakest ($h = 368$) and 1.81× at its strongest ($h = 8$) — which is the direction trunk-sharing predicts, and at $h = 100$ it is 1.65×, **71%** of the improvement. So the mechanism is supported and it is the larger part of the effect where the method operates. At the open-loop diagnostic horizon of $h = 368$ the split reverses — 57% of the improvement there is the ensemble simply predicting better — so a reader who takes the 2.53× figure at that horizon as a measure of the architectural effect would overstate it. We report both columns for that reason. (An earlier draft gave the σ range above as the $h = 1$ and $h = 8$ values, which do not span it — $h = 368$ ’s 1.49× falls below the stated floor. Found by the horizon sweep, which flagged the sentence for carrying two horizons’ figures while naming two others.)

What this does and does not license. It licenses saying that **the released ensemble’s disagreement understates epistemic uncertainty partly because its members are not independent models**, with a measured size at the horizon that matters. It does not license attributing the whole gap to trunk-sharing: §11 sets out that independently-seeded runs differ in *both* initialisation and data ordering, so this comparison **bounds** the architectural effect rather than isolating it, and the bound is generous to the mechanism by construction.

And it does not repair the interval. The independent ensemble is 5.2×

 overconfident at $h = 100$ with 15.31% coverage where a calibrated Gaussian gives 68.27%. Better by a factor of 2.03, and still not an interval. Building the ensemble properly is worth doing and it is not sufficient; §6.8’s per-horizon multiplier remains the only thing in this paper that restores nominal coverage.

7 Defects in the released pipeline

7.1 Ten unmarked episode boundaries. §3. The window builder reads a termination column that is identically zero, so it marks all 9,961 windows valid.

7.2 Training and evaluation disagree on action alignment, and evaluation is the broken one. Row t holds the action that *produced* state t . The training path pairs states and actions index-for-index, which is causally correct. The evaluation path feeds the action from $t-1$ to predict state t — stale by one step. Scored correctly the released checkpoint is materially better than its own released evaluation reports: nRMSE at $h = 368$ falls from 1.3228 under the released pairing to 0.7572 under the causal one, so the released evaluation overstates its own model’s error by 75%.

7.3 No held-out evaluation. Evaluation trajectories are drawn from training data. For the released checkpoint, trained on the entire file, no held-out measurement is possible at all.

7.4 What the spliced windows cost: nothing measurable. We trained a contaminated arm on 7,882 windows — the clean 7,687 plus 195 splices — and, because that confounds *content* with *count*, a duplication control adding the same 195 windows as exact copies of windows already present.

The arm’s contamination rate is 2.47%, against the reference pipeline’s 3.53%. It is deliberately lower: we splice only the 5 boundaries whose *both* sides are training episodes, because 4 of the 9 put held-out rows into training. That is a leakage problem rather than a physics one, and including it would have invalidated our own comparison. So this experiment measures the cost of training on physically impossible transitions, and not the reference’s full exposure.

Training loss over the final 250 iterations: duplication costs 0.90%, splicing costs 21.57%. The bootstrap interval on duplicated – clean is $[-0.0061, 0.0215]$, including zero. So the rise is caused by splice content, not by dataset size — a control we ran only because the first version of this finding inferred the mechanism without it.

In rollout, across 32 cells (two arenas $\times$ two trajectory lengths $\times$ two checkpoints $\times$ two horizons $\times$ two metrics), contamination hurts in 0 of 32 and helps in 9 (Figure 5a). The control is inert, differing from clean in 2 cells.

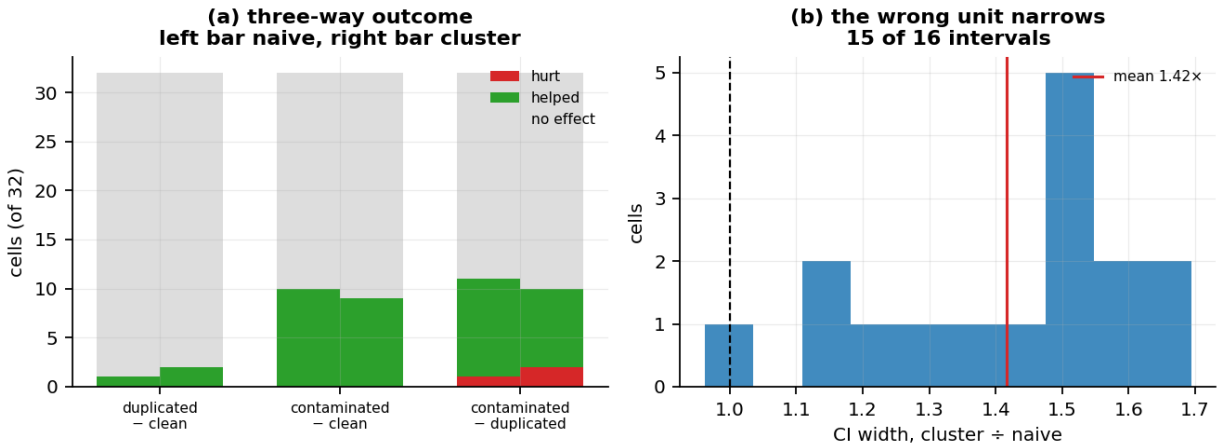


Figure 6: The contamination control. (a) outcome across 32 cells for each arm pair, naive bootstrap on the left of each position and cluster bootstrap on the right; the duplication control is inert. (b) distribution of the ratio of cluster to naive confidence-interval width, with the mean marked. Resampling trajectory-step pairs rather than whole trajectories narrows every interval.

"Costs nothing" is the wrong summary, and we should not use it. Splicing raises training loss by 21.57% against duplication’s 0.90%, and improves rollout in 9 of 32 cells. Both are measured effects in opposite directions, which is the signature of regularisation: the spliced windows contain transitions the model cannot fit, it fits the rest less tightly as a result, and it rolls out slightly better. The defensible statement is that **at this rate the splices do not harm rollout and appear to help slightly**, not that they cost nothing. **The unmarked boundaries remain a real defect on leakage grounds; what is now measured is that the physically-impossible-transition component costs nothing detectable at this rate.**

7.5 The released artifacts do not reproduce the released checkpoint’s variance state. The σ collapse is linear in iteration count and its rate is nearly identical across our runs (§6.3), which makes it a clock. Extrapolating it to the released checkpoint’s σ state implies **153,270** optimisation steps at the configured learning rate — against a released configuration that says 500, a paper that says 2,500 and a checkpoint tagged 5,000. A second parameter on a slower gradient path implies the same order independently. The defensible claim is narrower than it first looks, and narrower than an earlier draft of this paper made it: **no constant-rate run from the released initialisation at the configured learning rate reaches that variance state at any of the three stated counts.** A warm start or a different `log_delta_logstd` initialisation would explain it with no inconsistency at all, and we can exclude neither. The first author’s account is that the released repository is several revisions removed from the setup that trained the checkpoint, which supplies exactly such a mechanism. So this is a **documentation gap between a release and a**

run — common, worth recording, and much less interesting than an inconsistency. Appendix H gives the arithmetic and the five assumptions it rests on, because it is what let us detect the gap at all.

8 Method

An append-only ledger. Every claim here has a permanent identifier, an evidence class (source, data, run, external, inference) and a status, in `FINDINGS_LEDGER.md` (244 entries). Claims are never edited in place: one that turns out to be wrong is marked superseded, pointed at what replaced it, and kept.

Pre-registration, and one failure of it. Decision rules were committed to git before the data that tested them, with one exception. Figure 4 gives the lead time for 8 of them and Appendix G for all 15, every one of which now carries one; 7 of Figure 4’s are positive and 1 is not. Figure 4 plots the set it was drawn over and is not re-drawn: the rules this revision adds have lead times of minutes to hours and would render as slivers. Every positive bar is a difference of two commit timestamps. **The negative one is not, and the difference matters:** it is the duplication-control rule (§7.4), whose *data* side is the moment the control runs finished, and that is a line in `results/control_driver.log` rather than a commit. The log records wall clock with no date and no offset, so both are taken from the commit that introduced that line — which makes the figure reproducible outside this machine’s timezone, and it was not: the same arithmetic gave a different answer in every timezone until the offset stopped coming from the reader’s clock. The rule was stated in conversation before the runs and reached git **2.9 hours after they finished**, and we found it only by auditing our own `git log`. The measurement stands — the arm was built without reference to its outcome — but the claim that it was pre-registered does not, and we withdraw it. A discipline that is only checked when it succeeds is not a discipline.

Six retractions on our own evidence, out of 19 superseded claims kept in the record, plus six that withdraw framings rather than numbers (Appendix D lists them). The most consequential of those is **S-15:** the inference from per-dimension sign counts to a binomial P-value, which assumed an independence the 45 state dimensions do not have (§6.6). It was named by position here until the second pre-submission review entered four more framing retractions and moved it. Found by our own pre-submission audit, it withdraws the strength of evidence behind what an earlier draft called the strongest result here.

A statistic that was resampling the wrong unit. Our bootstrap pooled three seeds over a shared trajectory set and resampled the pooled vector while reporting the independent-trajectory count, so each trajectory appeared three times. Resampling trajectories instead widens intervals by a mean 1.42× and changes 1 of 16 verdicts, in an $h = 8$ cell already recorded as unresolvable. Every long-horizon verdict survives; both units are reported.

Reproducibility, and a build that checks its own prose. `./reproduce.sh -quick -force` regenerates 47 artifact files and 8,186 numeric values from a clean clone: 7,981 bitwise identical (97.50%), 27 equal to within the verifier’s floating-point tolerance but not bitwise, and 178 differing. Those three account for the 8,186 exactly. A further 25 values are present in the committed artifacts and absent after regeneration, across 3 files; they are counted separately because a value that is not produced twice cannot be compared twice. **The claim is narrower than the percentage makes it sound, and we would rather state its size than have a reader derive it.** A clean clone already contains every committed artifact, so the only honest test is the subset the run actually rewrites: 8,186 values, or 0.90% of the 905,391 numeric values under `results/`. The other 897,205 are carried in, prove nothing about reproduction, and are never folded into the figure — an earlier version of this section counted them and overstated the result by about fiftyfold. **What "every numeral" means is itself checked.** A paper cannot substitute a section number or an arXiv identifier, so the claim is partitioned: every *measurement* is substituted, and each of the 582 numerals that is not one is classified as an address, a horizon label or a declared constant — 18 classes and 23 declared exceptions, with the build failing on anything left over (`results/typed_numerals.json`). That audit exists because the abstract used to claim no number here was typed, which was false; the count was printed on every build and asserted by nothing. Verifying that every numeral came from an artifact says nothing about the sentence built around it — six defects in an earlier draft were of exactly that kind, all downstream of correct numerals. The build therefore also verifies **52 comparative claims** across 21 kinds, each pinning a fragment of the paper’s own text *and* a relation recomputed from the artifacts; all pass, and each is run

against a deliberately corrupted expectation on every build and must fail, 52 of 52 caught. **Appendix D gives the argument, the kinds, the self-test, the four defects the self-test has found in the checker itself, and the two exclusions from the numeric comparison.**

9 Actionable lessons

Six things a practitioner can apply without reading the rest of this paper.

Use ensemble disagreement as a ranking signal, but price it against the free alternatives first. Do not read it as a distance. And expect it to degrade with horizon. At one forecast step it ranks whole rollouts almost perfectly — $+0.994$ [$+0.918$, $+0.999$] across the 20 trajectories, and that is a ranking of *rollouts*, not of moments within one, because at $h=1$ there is only one moment. Over the full rollout it falls to $+0.605$ [$+0.545$, $+0.694$]. That decay is the useful part: the signal is excellent where you can check it cheaply and merely good where you most need it. It still beats the free alternative — the forecast step index — at every horizon we tested, on a paired test that excludes zero at 5 of the 5 horizons where the index is defined — every one of them — and it retains $+0.596$ once that index is partialled out (§6.7). **What it does not clearly beat is the model’s own predicted step size**, which costs nothing and needs no ensemble at all: that quantity ranks error at $+0.4697$ against disagreement’s $+0.6053$, and the margin between them, $+0.1357$, is below the 0.2891 this sample can resolve. Disagreement does carry information the subtraction does not — it retains $+0.5430$ once step size is partialled out — but a practitioner about to pay for five members should price the subtraction first. That is a real signal, not a re-encoding of how far ahead you are looking — and not merely a report of which episode is hard: with both the forecast depth and the rollout held constant it still correlates $+0.419$ [$+0.318$, $+0.576$] with error (§6.7). But it is too small to be an interval by a wide margin — at $h = 100$, the horizon the method itself rolls out over, $33.4\times$ [28.7 , 39.0] on the released checkpoint and $10.5\times$ [9.0 , 11.5] on the ensemble-5 arms we trained — and a risk gate or safety margin that reads σ as a distance is not supported at any horizon, on either.

If you need the interval, rescale per horizon, not globally. One multiplier per forecast horizon, fitted on held-out data, brings coverage within 10 points of nominal on every held-out cell; a single global multiplier manages 2 of them (§6.8). The held-out cells are 6 horizons $\times$ two fold directions on the same 4 trajectories, not independent trials, so read the sweep as consistency and the per-cell deviations as the evidence. The fitted multipliers span $9.31\times$ across horizons, which is precisely why one number cannot serve.

Do not convert per-dimension sign counts into P-values. State dimensions in a robot are physically coupled and share a forecast-depth trend, so an independent-trials null is badly wrong — in our tables by up to $10^{13}\times$ (§6.6). Permute whole trajectories instead. We shipped the binomial version in an earlier draft and it made our weakest evidence look like our strongest.

Count independent trajectories, not trajectories. Two 400-step windows that overlap at all are one piece of evidence, not two. The held-out arena here contains 4 independent 400-step trajectories however many windows are drawn from it, and that number — not the window count — bounds every long-horizon claim. Reporting an interval beside a trajectory count rather than an independent-trajectory count overstates precision, and resampling pooled seed $\times$ trajectory values instead of trajectories narrows intervals by a further $1.42\times$ (§8).

Anchor a decision rule to the horizon the claim is about. Our first pre-registered rule was anchored at $h = 8$, the training forecast horizon, and returned "cannot be settled". The claim was about deployment horizons. The rule was correct in form and pointed at the wrong regime, which is a failure mode that pre-registration does not protect against on its own.

Check that the implemented loss is the described loss before reproducing any number from it. The paper describes two loss terms; the implementation has 7. The predicted variance has an optimum at zero under the implemented one, which is why the released checkpoint’s σ is $11,683\times$ smaller than its own error at $h = 100$, and $20,669\times$ at $h = 368$. Reading the loss took an afternoon and explained a result that would otherwise have looked like a training bug.

10 Broader impact

This is a reproduction of a dynamics model on public simulation data, and the reproduction itself carries no significant risk of harm: no new capability, no personal data, no deployment.

The finding does bear on safety, in one specific way worth stating. The distortion is not a units problem a tuned coefficient absorbs: the constant that would best rescale this σ grows by a factor of 4 between one step and 368 (§6.2), so a penalty weight chosen for one rollout depth is wrong at another. The method this paper examines uses its uncertainty estimate as a **trust metric** — a reward penalty that steers a policy away from states the model is unsure about. That use is supported by our measurements, with one qualification a deployer should weigh: a free signal — the model’s own predicted step size — ranks error nearly as well, +0.4697 against +0.6053, so the ranking evidence alone does not establish that the ensemble is worth its cost (§6.7). But a downstream user who reads the same quantity as a *calibrated interval* — a safety margin, a confidence bound, a gate on when to hand control to a fallback controller — would be materially misled: at the horizon the method itself rolls out over — $h = 100$ — the released checkpoint’s ensemble disagreement is 33.4× smaller than the realised error, giving 4.61% coverage where 68.27% is expected. On hardware, a margin that is wrong by that factor is the difference between a conservative controller and one that believes it is safe when it is not.

We think that makes the finding worth publishing rather than the reverse, and it is the reason §6 reports coverage rather than only correlation.

11 Limitations

Effective sample size bounds every long-horizon claim. The out-of-sample arena has 4 independent 400-step trajectories. That is the binding constraint on §5, and no amount of trajectory oversampling changes it.

Ensemble size — no longer an open question, but not a closed one either. Our main experiment runs at ensemble size 1, where the epistemic term is identically zero by construction, so every epistemic measurement here was originally made on the released checkpoint alone. We since trained 3 Arm A arms at ensemble size 5 (§6.7). They reproduce the *direction* of §6.7’s finding in 12 of 12 seed-horizon cells and the *calibration* failure at 10.5× at $h = 100$ (13.0× at $h = 368$) — but the pre-registered rule governing the replication returns **DOES NOT GENERALISE**, because its second condition needs the paired difference to exclude zero at a majority of horizons and it does so at 1 of 4. The binding constraint is the same one this section opens with: our arms have a genuine held-out arena of only 4 independent trajectories, and the rule was written without checking what it could detect there. **So §6.7’s finding is established on the released checkpoint and supported but not established on a model we trained.**

One dataset, one gait, one terrain. All commands are drawn from one bounded box and the gait is a single trot throughout. "Generalisation" here means across velocity commands, not across gaits or terrain.

The per-horizon recalibration is fitted and tested on two episodes only. §6.8’s remedy transfers across the two held-out episodes in both directions, which is the strongest test the released split allows, but two episodes is not a demonstration that the multipliers transfer to a new robot, gait or terrain. Treat the lookup table as a recipe to refit, not as constants to copy.

Two secondary analyses rest on a single training seed — the long-horizon trend fit and the per-dimension matched comparison, both computed on seed 1 alone. The headline A/B result is not among them: it is a three-seed mean with per-seed values reported (§5). This is recorded in the artifacts themselves.

The ranking claim is not established as needing an ensemble. §6.7 shows ensemble disagreement ranks realised error well and survives six controls. It does not show that the ensemble is necessary for that: under a rule committed before the comparison (M-51), the magnitude of the model’s own predicted state change — a subtraction, requiring no ensemble and no second model — ranks error at +0.4697 against disagreement’s +0.6053, and the margin between them is smaller than this sample can resolve. Disagreement retains +0.5430 with that baseline partialled out, so it is carrying information the subtraction is not; what

is open is whether that increment is worth five models. Settling it needs more independent trajectories than the released dataset contains, and it is the cheapest open question here for anyone with a second dataset.

We did not measure what the miscalibration costs. We show that the penalty the follow-up applies is miscalibrated as a scale — $33.4\times$ overconfident at $h = 100$, the horizon its own imagination rollouts run to — but the only use the method makes of that quantity is to shape policy learning, and we did not train a policy. A miscalibrated scale that enters as a relative penalty across candidate actions may cost little, or may cost a great deal; our measurements cannot distinguish those. **The finding bounds what the quantity reports, not what it costs.** That distinction is easy to lose and we do not want a reader to take the ratio as a measure of harm.

The per-dimension ordering tests are underpowered at every sample size we can reach. Once the coupling between state dimensions is respected (§6.6), the out-of-sample arena’s 4 independent trajectories admit a smallest attainable P-value of 0.04167 — coarser than the multiplicity-corrected threshold 0.001667, so that arena cannot reject at any effect size whatever. The larger arenas can reject and do not: over all ten episodes the smallest P in the family is 0.0037 against a threshold of 0.001667. Resolving it at $h = 368$ needs more episodes than the released dataset contains, rather than a better test. **At $h = 128$ and below that is no longer true and we say so:** a shorter evaluation unit gives 14 independent units at $h = 100$ where the 400-step unit gives 4 (M-64), and we did not rerun this permutation family there. Note the scope: this limits the *per-dimension* evidence. The aggregate scalar the method applies is separately and more strongly supported (§6.7), on the same trajectories, because it is one test rather than forty-five coupled ones.

No family-wide correction is applied across our own pre-registered rules. There are 15 of them with per-rule verdicts (Appendix G) and we report each against the thresholds it was committed with, not against a corrected family threshold. Pre-registration is what licenses that: each rule is a separate question committed before its data, not one search over many outcomes, and a rule that fails is reported as failing. A reader who prefers the corrected reading should apply it; we state the count so that is possible.

The independent-ensemble comparison bounds the trunk-sharing effect rather than isolating it, on three axes. §6.10’s contrast trains five models at five seeds and scores them together. Independently-seeded runs differ in **both** initialisation *and* data ordering, whereas the shared-trunk heads differ only in head initialisation. They also differ in **capacity**: the independent arm carries 3,570,820 state-pathway parameters against the shared-trunk arm’s 1,024,132, a factor of 3.49, because each member brings its own trunk. Greater capacity can inflate σ as well as shrink error, and σ is the column the mechanism claim rests on — §6.10’s decomposition separates the σ gain from the accuracy gain, but it does not separate capacity from independence. **Capacity is no longer one of them.** M-49, committed with its minimum detectable effect before any of its models existed, trains 5 independent members at `rnn_hidden_size` 124 against the released 256, giving 1,023,880 state-pathway parameters against the shared-trunk arm’s 1,024,132 — a ratio of 0.9998, where §6.10’s original contrast carried 3.49. **With capacity held fixed the independent ensemble is still better calibrated on every shared-trunk seed, every paired interval still excludes zero, and the coverage gain of +6.42 points still clears its own MDE.** The effect does not vanish when the confound is removed.

It does shrink, and by more than this design can resolve. The overconfidence improvement falls from $2.03\times$ unmatched to **1.79 $\times$ matched**, against an MDE of $2.00\times$. M-49 therefore returns **UNDER-POWERED** — 5 of its 6 conditions hold and the ratio threshold is the one that does not. That is the third branch the rule names, and it names it because its MDE was almost exactly the size of the effect it re-tested; §6.10’s own text said so before the runs rather than after them. So: trunk-sharing is **not** explained away by capacity — the effect is in the same direction on every pair with every interval excluding zero. Whether capacity accounts for *any* of it is a different question and this design does not answer it: the point estimates fall from $2.03\times$ to $1.79\times$, a difference of about 0.24, which is untested — no artifact here pairs the two independent arms against each other — and far below the $2.00\times$ this comparison can resolve. An earlier draft of this sentence said capacity accounts for some of the effect; that is more than R-73 supports and it is withdrawn. Closing it needs more independent trajectories than the released dataset contains. The comparison still conflates trunk-sharing with data-order diversity, which M-49 does not address and this paragraph does not claim it does. That asymmetry is deliberate and it is generous to the mechanism: if the

overconfidence factor barely moves despite the handicap, the finding is strong in the direction of *architecture is not the explanation*; if it moves a great deal, the design flaw is identified but not cleanly attributed to trunk-sharing alone. Isolating it would need an ensemble that shares data ordering and not parameters, which is a different experiment. M-44 states this in its own text, committed before the runs.

§6.4’s mechanism is a structural fact plus a hypothesis, and the two are separable. That the five members share a trunk, a hidden state and 89.15% of each member’s parameters is measured, from source and from the checkpoint’s tensors. That this *causes* the epistemic miscalibration is the hypothesis, and only §6.10 bears on it.

Deliberately out of scope, and stated so a reader does not assume otherwise. No policy-learning result of either paper is tested — no simulator, no RL loop, no ANYmal, and no policy is trained anywhere in this work. The sample-efficiency comparison (roughly 6M against 250M transitions) is not tested for the same reason. Nothing here uses a GPU. And **we did not test whether the $\sigma = 0$ optimum affects other descendants of the PETS parameterisation (§2):** the parameterisation is inherited line for line and the objective is not, which makes the hypothesis well-founded and untested. Testing it needs other repositories, and we make no claim about them.

We did not reproduce the policy-learning results of either paper. This is a dynamics-model reproduction only.

12 Conclusion

The Robotic World Model’s central training claim reproduces, and the margin is large. Neither uncertainty output of the follow-up that adds them reports what a reader would take it to report. At $h = 100$, the horizon the method’s own imagination rollouts run to, the aleatoric σ is $11,683\times$ smaller than its own error, and the cause is that the objective’s optimum is $\sigma = 0$ with the term that should prevent this cancelling out of the gradient. The epistemic term the method actually penalises with is better by a factor of 349 and still $33.4\times$ [28.7, 39.0] overconfident where it is used — both figures at $h = 100$.

The more useful finding is asymmetric, and it cuts both ways. The scale failure is established and large — but it is repairable: a per-horizon multiplier, fitted on one held-out episode and scored on another, restores nominal coverage on every held-out cell where a global multiplier restores 2 of them — 6 horizons in each of two fold directions, on the same 4 trajectories, so not independent trials. And the ranking use the follow-up claims does survive a real test: against the forecast step index, a free baseline neither original paper ran, ensemble disagreement wins at every horizon and keeps +0.596 once the index is partialled out. Against a second free baseline it does less well: the model’s own predicted step size ranks error at +0.4697 against disagreement’s +0.6053, a margin this sample cannot resolve, so the verdict is SURVIVES entry-res ONLY. **The control this rests on is the one that removes trajectory difficulty rather than forecast depth:** with both the rollout and the depth held constant, disagreement still correlates +0.419 [+0.318, +0.576] with realised error (§6.7, M-45). That is a smaller number than the +0.605 pooled figure and it is the one that means what a practitioner needs it to mean — so it is not a re-encoding of the clock, and not merely a report of which episode is hard. That is the closest either original work comes to a claim this reproduction strengthens rather than qualifies — and even there the strengthening is of the ordering, not of the ensemble that produces it, since a free subtraction ranks nearly as well.

What does not survive is the per-dimension form of the ordering evidence. Three of the five σ estimates we measured order their own errors better than chance in direction — the epistemic term on every one of the 45 dimensions at $h=368$, and the faithful and teacher-forced arms. That count is a direction, not a tally of independent trials — the dimensions are physically coupled, and the permutation test over whole trajectories is the statistic (§6.6). The released checkpoint’s *aleatoric* head does the opposite, ranking error inversely at $h = 368$ on every one of 45 dimensions over all ten episodes and at chance on the held-out pair alone — a dependence on arena that §6.6 sets out. The corrected arm sits at chance in both. And once the physical coupling between state dimensions is respected by permuting whole trajectories, no per-dimension count in this paper reaches significance after multiplicity correction. We report that rather than the independent-trials P-values an earlier draft carried, which were wrong by up to a factor of about 10^{13} on the cells we

had cited as evidence. Neither quantity yields a usable interval. Uncertainty in this family of models should be read as a weak ordering at best, or fixed at the objective; it should not be read as a scale, and a ranking use deserves its own validation on the deployment distribution rather than trust inherited from here.

13 Data and code

The full repository — code, every artifact under `results/`, and `FINDINGS_LEDGER.md` with the complete claim record including the retractions — accompanies this submission as anonymised supplementary material, and will be released under a permanent archival identifier on acceptance. Neither upstream repository is redistributed; `setup.sh` fetches both at pinned commits and verifies two SHA-256 hashes.

The repository’s history was rewritten once, and §8 depends on that history, so we say what changed. A supplementary file quoting private correspondence was committed and briefly published before consent to quote it had been given; it was purged from the history rather than merely deleted, because a deletion commit leaves the content recoverable from a public repository indefinitely. Purging a path rewrites every commit from the one that introduced it onward, so **14 of the commits Figure 4 cites keep their identifiers and two do not** — the two whose data post-dates that file. Timestamps, content and ordering are unchanged; only the hashes moved, and Figure 4 resolves each rule by its commit subject for that reason. The transcript itself reaches reviewers in the anonymised supplementary archive, which is not published.

The pre-registration argument in §8 rests on commit timestamps, and those are author-settable via `git commit -date`. That matters, because §8 is load-bearing. Two things address it. The supplementary material includes an anonymised `git log` covering every commit cited here, so the ordering in Figure 4 is checkable at review time. And **the repository was archived by a third-party archive before submission**, under a permanent identifier whose visit timestamp is not author-controllable. Neither the identifier nor the date of that visit appears here: both resolve to a named repository, and a date is a one-field lookup away from an origin. They are disclosed on acceptance.

What that archive establishes should be stated precisely, because it is easy to overclaim. It does **not** prove any individual commit date is genuine. It proves that the repository, with the whole pre-registration history in the form this paper cites, existed no later than that archival moment, as recorded by a third party with no interest in the claim — so nothing in the record can have been back-dated afterwards. That bounds §8 rather than proving it, and a reviewer should read it as such.

On anonymity, stated rather than implied. The code and data for this work are public, as they are for most reproducibility work, and a reviewer who chooses to look can identify the author. The submission is anonymised — the bundle is scrubbed and asserted clean of a deny-list, and the files that carry identity are excluded from it — but that is anonymity of the *submission*, not unfindability of the work. Making the repository private would remove the identifying link and also remove the checkability §8 depends on, which is the worse trade. The decision and its reasoning are recorded in `docs/DOUBLE_BLIND_DECISION.md`.

14 References

1. C. Li, A. Krause, M. Hutter. *Robotic World Model: A Neural Network Simulator for Robust Policy Optimization in Robotics*. arXiv:2501.10100v1, 17 January 2025. *Read at v1, whose Roman-numeral sectioning our references follow; v2 (23 April 2025) renumbered to Arabic and moved IV-C into Appendix A.4.1.*
2. C. Li, A. Krause, M. Hutter. *Uncertainty-Aware Robotic World Model Makes Offline Model-Based Reinforcement Learning Work on Real Robots*. arXiv:2504.16680v1, 23 April 2025. *Read at v1; now at v3, last revised 8 Jan 2026. §5.1 and Eq. 4–5 keep their numbers there; every figure and appendix table has moved, and the model is renamed RWM-O to RWM-U — the same model, with the letter re-expanded from "Offline Robotic World Model" to "Uncertainty-Aware Robotic World Model". The two names never co-occur: v1 uses RWM-O 39 times and no RWM-U, v3 the reverse. The crosswalk is in `results/original_paper_figures.json`.*

3. Z. Abbas, S. Sokota, E. J. Talvitie, M. White. *Selective Dyna-style Planning Under Limited Model Capacity*. ICML 2020. arXiv:2007.02418, 2020.
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14. A. Malik, V. Kuleshov, J. Song, D. Nemer, H. Seymour, S. Ermon. *Calibrated Model-Based Deep Reinforcement Learning*. ICML 2019 (PMLR 97:4314-4323). arXiv:1906.08312, 2019.
15. Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, J. Snoek. *Can You Trust Your Model’s Uncertainty? Evaluating Predictive Uncertainty Under Dataset Shift*. NeurIPS 2019. arXiv:1906.02530, 2019.
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17. Y. Wen, D. Tran, J. Ba. *BatchEnsemble: An Alternative Approach to Efficient Ensemble and Lifelong Learning*. ICLR 2020. arXiv:2002.06715, 2020.
18. T. Yu, G. Thomas, L. Yu, S. Ermon, J. Zou, S. Levine, C. Finn, T. Ma. *MOPO: Model-based Offline Policy Optimization*. NeurIPS 2020. arXiv:2005.13239, 2020.

Entries 3–18 are the §2 bibliography, generated from `results/t1_bibliography_verified.json` rather than listed here — a hand-maintained list of what a paper cites drifts exactly as a hand-typed count does, and this one had: six entries cited in §2’s prose appeared in no reference entry while the note below claimed all of them verified. Each was checked against the paper itself: title, full author list and venue from the arXiv record, and for any sentence this paper attributes, the sentence matched verbatim against that paper’s own text — 16 of 16 entries and 17 of 17 attributed fragments (`results/t1_bibliography_verified.json`).

A Verification chain

What every downstream number rests on. Each level was passed before the next was attempted.

level	claim	result
shapes	parameter counts match the reference	exact
wiring	inference outputs match the reference module	0.000e+00 , bitwise
indexing	the harness feeds the actions it claims	bitwise against the raw CSV
residual	the zero-delta model is the hold-last floor	1.192e-07
objective	losses and gradients match	0.000e+00 across 7 terms, 106 tensors
trainer	can memorise a single batch	1,506× loss reduction

B Reproducing

```
./setup.sh # clone upstreams at pinned commits
python3.11 -m venv .venv && . .venv/bin/activate
pip install -r requirements.txt
./reproduce.sh --quick --force # everything except training
```

`-force` matters: a clean clone already contains each stage’s declared output, so without it every stage skips.

Runtime. Training stages are excluded by `-quick`, which is what makes the quick path practical. Training all 31 runs takes **48.2 hours** of recorded wall clock on two CPU cores: 19.7 hours for the 6 runs at 10,000 iterations and 28.5 for the remaining 25 at 2,500. (Those were rounded to whole hours in an earlier draft, where 20 + 27 did not make 46; the `arithmetic` check now asserts that a stated total equals the sum of its stated parts.) The longest single run is 4.4 hours. An earlier version of this appendix said 22 hours; that figure predated the 6 ten-thousand-iteration runs added for the three-seed headline, and is corrected here from the `wall_clock_s` field of every run artifact rather than re-estimated.

5 of those 31 runs, 1.9 hours, are M-49’s capacity-matched arm at `rnn_hidden_size 124` rather than the released 256. They are part of this project’s CPU spend and are counted in the total above; they are **not** part of the 26 runs §6.3 fits the σ -collapse rate over, because that rate is a property of one architecture and mixing widths into it would make "nearly identical across runs" a claim about two different models. Every run artifact records the width it trained at, and `paper_numbers.py` selects the collapse family by that field rather than by filename — it did neither until the first capacity-matched run walked into the family through a glob. **That fix reached `paper_numbers.py` and `paper_figures.py` and not every script:** a second unguarded glob survived in the ensemble-5 comparison until this revision found it, moving two published values (M-66). A sweep of all 21 pattern-based input discoveries in `scripts/` and `src/` now classifies each as an open population or frozen at write time; 1 was frozen, and artifacts that discover their inputs now record the file list they were computed over.

C Verifying the paper’s own claims

The six numbered retractions, in order. In order: a premise about forecast decay that turned out not to exist in the code; a framing of the released checkpoint as "clearly informative" that rested on an $n=10$ estimate we ourselves showed to be biased low; an aggregation artifact that inverted a published-model comparison in our favour, withdrawn when the gating checks we had written refuted it; a per-dimension comparison that turned out to be unmatched; the claim that σ is input-independent "in all four models", made against a table holding three; and the phrase "the released checkpoint’s uncertainty output", singular, when the checkpoint emits two and we had measured the one the method discards. **The six framing retractions**, withdrawn as stated claims rather than as numbers, and generated from the ledger rather than listed here — a typed enumeration beside a generated count is the same defect as a typed count. This one was typed with two entries at a time when the ledger held two, and the revision that added four more replaced it with a generated list in the same commit, so it never actually stood wrong. It appears here as

a note rather than above as an entry for that reason, and the reason is luck: nothing compared the typed enumeration against the count beside it, and had the two changes landed in separate commits the paper would have said six and enumerated two:

- **Task 3’s duplication rule was pre-registered (S-12).**
- **The binomial P-values attached to every dimension count (S-15).**
- **The two largest held-out deviations are in opposite directions (S-16).**
- **The eight untested claims are, without exception, about policy learning or hardware (S-17).**
- **The per-member σ is worse by three orders of magnitude (S-18).**
- **The released checkpoint cannot have come from the released recipe (S-19).**

The last four were entered by the second pre-submission review, in a single commit — which is how that count is established rather than recalled. S-16, S-17 and S-18 are sentences of the 24 August draft that were false. S-19 is different in kind and worse in one respect: §7.5 had already narrowed that claim in the paper, and the narrowing was never entered in the ledger, so the withdrawn version went on standing in the ledger’s own contributions summary and in the public README after the paper had withdrawn it. A retraction that holds in one document and not in the repository is not a retraction, and the `retraction-consistency` check now reads all five reader-facing files rather than three. Each is an entry in `FINDINGS_LEDGER.md` with its evidence and its successor.

`build_paper.py` asserts that every printed number came from a named artifact. That is a guarantee about *provenance*, and it is silent about *relations between* provenanced numbers. Six failure modes survive it, and all six occurred in this paper. Five are relations between provenanced numbers; the sixth is not a relation at all and is set out after the list:

- **an interval relation that is not the one asserted** — "the intervals do not overlap", where at $h=128$ they overlap across 0.604–0.643. **This one has now been wrong twice.** The correction said the distinction mattered "at exactly one place"; adding $h = 100$ to the grid made it two, and the sentence recording the first error carried the second. A stated *frequency* — "at exactly one place", "in all four", "the only" — is a claim about a count, and no kind bound one to a recomputed count until *frequency-consistency*;
- **an extremum that is not the extremum** — the worst-calibrated held-out cell named as epistemic at $h=1$, which is third; the largest deviation is aleatoric at $h=100$. **This one has now been wrong twice as well**, and the second time was here rather than in §6.8. This sentence named $h=128$, which was the extremum before $h = 100$ entered the evaluation grid; §6.8 was re-derived when the grid changed and the sentence describing the *correction* was not. Both now read the same key, and the *extremum* kind covers this appendix and not only the section that computes it;
- **a stated change with the wrong sign** — "a change of +0.010", where partialling the forecast index out *reduces* the correlation;
- **two prose descriptions of one ratio that disagree** — "nearly three orders of magnitude" in the abstract against "two orders" in §12, of the same $600\times$ at $h = 368$;
- **a count attributed to the wrong evaluation arena** — 0 of 45 over all ten episodes asserted where the table beside it printed the held-out arena’s 20 of 45.

None is a numeral. None appears in `results/paper_numbers.json`. Each was typed.

A sixth failure mode is not a relation at all, and it defeated the gate rather than evading it. `build_paper.py` asserted that no `{{`-delimited placeholder survived substitution, and none did — while a sentence of §6.7 reached the PDF as an empty one-column table. The sentence contained `|r_dd|`, the line wrapped so that the pipe began a line, and the Markdown-to-LaTeX converter read a leading pipe as a table row. Every numeral in it was correct and provenanced. The gate now refuses three further shapes as well as unresolved braces: a pipe-led line with no separator row beneath it, a single-braced token that names a real key, and any key resolving to an empty or null value. The converter requires the separator row before it will build a table.

The machinery, and the two lines that matter about it. `scripts/check_comparative_claims.py` verifies 52 claims across 21 kinds, each pinning a fragment of this paper’s text and a relation recomputed from the artifacts, and each run against a deliberately corrupted expectation on every build so that a check which can no longer fail is caught. The registry, the self-test’s mechanics and the Four defects it has found in the checker itself are in `docs/BUILD_CHECKS.md`, shipped as supplementary. **The evidence that any of it is load-bearing is one sentence:** the comparative checks caught three defects in text written during this revision — two calibration figures that named no horizon, and one numeral quoted three ways in a single sentence — none of which a human reader had noticed. **Two exclusions from the numeric comparison,** on the same principle in both cases: the number measures the machine, not the model. A third category is not an exclusion but a partition, and it bounds everything in this appendix: of the 905,391 numeric values under `results/`, a clean clone regenerates 8,186 and carries in 897,205. The reproducibility claim covers 0.90% of the directory and is silent about the rest. We state that fraction because a reviewer who computes it and finds we did not will reasonably discount everything around it.

The CPU budget. 5,004 timing fields and the 592 values of `results/step4_5_timing.json` — projected runtimes for configurations we did not run, peak resident memory, and the standard deviation across repeats. It cannot reproduce bitwise on another machine, or on this one under different load, and it records that about itself: across its 4 configurations the standard deviation of seconds-per-iteration across repeats runs from 5% to 32% of the mean (`ens1_bs256`) — on one machine, within a single measurement session. **Those three figures are themselves host measurements and we mark them as such,** since a sentence arguing that host-dependent numbers should not be printed as results cannot quietly print three of its own as though they were stable. On a second host the same run reported a different spread on a different worst configuration. Read them as one machine’s account of its own variability, not as a property of the code.

One of the build’s own gates fails, and we report it rather than retire it. The clean-clone check in `part_f_gate` requires that *no* regenerated value differ. 178 do, so the check fails, and it is published as failing. We did not give it a tolerance. The differences are concentrated in two artifacts — a stochastic dilution study whose values move in the tenth significant figure, and an index that stores line numbers in this document, which move whenever the document is rebuilt — but "concentrated in two artifacts we believe are benign" is a reason to read the check’s output, not to move its threshold. A gate that passes because its criterion was relaxed tells a reader strictly less than one that fails and says where.

One wall-clock-bounded diagnostic. `results/step4_4_overfit_ens1.json` stops after 2,700 seconds rather than at its 2,000-iteration cap, so it reaches a different iteration count on every machine — three different values across the three hosts we have run it on. Its iteration count and terminal losses are therefore a property of the host, and we do not quote any of them here: a number the build declares host-dependent has no business being printed as a result. **Its sibling from the same script is not excluded:** that run reaches its cap, and reproduces bitwise. Excluding by filename rather than by stopping rule would have dropped the reproducible one along with it, so the verifier decides from the artifact — a run that stopped short of its own cap was time-bounded.

D What testing the untested claims would require

§4's table marks 4 claims tested and the rest not. "Not tested" is an apology unless it comes with a price, so here is what each would cost. We give compute orders where we can estimate them honestly from this project's own measurements and say so where we cannot.

Everything below needs what this reproduction did not have: a simulator. Our arms train on the released CSV, which is a recording. Every untested claim needs *interaction* — a policy acting in an environment and the environment responding — and that means Isaac Lab, which needs an RTX-class NVIDIA GPU. No amount of CPU substitutes: the reference's data generation is GPU-parallel simulation, not a data-loading problem.

untested claim	what it needs	order
Sample efficiency, 6,000,000 against ~200M transactions (§V-E)	Isaac Lab, an RTX-class GPU, the MDPPO-PPO loop, and a PPO baseline run to convergence for the comparison	the reference reports 6,000,000 pretraining transactions and 50 min of RWM training on their hardware; the PPO baseline's 200M is the dominant cost
MDPO-PPO beats SAC and DDPG (§V-E)	the above, plus SAC and DDPG implementations at matched budgets	three policy-learning stacks, each tuned enough that the comparison is fair — the largest engineering item here
Zero-shot hardware transfer (§V-E)	all of the above, plus an NVSim, a safe test area, and the site-level stack	not estimable to compute; the budget constraint is hardware access, not GPU lease
Generality across quadruped, humanoid, manipulation (§V-D)	repeated state-action data from a humanoid and a manipulator, which means Isaac Lab and a policy in each environment to generate it — the released CSV is one robot on one terrain	our data-generation run per morphology, plus one world-model training run each at our 4k-2.5 scale; the model training is the cheap half and the data is not
Offline MRL on real robots (2004-2008v1) §5	a real robot, a logged dataset from it, and the offline MRL loop	not estimable to compute; hardware access again, and a claim the follow-up itself states as prospective
Whether the penalty improves the learned policy (2004-2008v1) §5	Isaac Lab, the MDPPO-PPO loop, and at minimum an ablation with the penalty weight at zero	one policy-learning stack; the cheapest of the four, and the one that would leave §11's open question about what the misalignment costs
Best MDP, RSSM, transformer baseline (§V-C)	no simulator needed — but the line release ships only the RSSM variant, so all three baselines would have to be implemented	comparable to our own modified 4k-2.5 of CPU training per architecture, times three, if run at our data budget
M/2, N/8 optimal (§V-C)	no simulator needed; a sweep over M and N at our data budget	our 31 runs took 48.2 h on two cores; a modest sweep is a small multiple of that

The two at the bottom are within reach of this setup — the M/N configuration sweep and the MLP/RSSM/transformer baseline comparison — and are the honest next steps for anyone extending this work on CPU. The six above them are not, and no amount of care with the released CSV changes that. This table has one row per untested claim; it listed 2 fewer than that until the assertion that counts its rows against Appendix F's was written, and the two it omitted were the two whose cost is hardest to state honestly.

What we would do first. The penalty ablation. It is the cheapest of the simulator-requiring items, it bears directly on the one limitation §11 states that our measurements cannot bound — whether the miscalibration we document costs anything downstream — and it needs no hardware.

E Every claim of the originals, and what we did with it

The body's §4 summarises this table. It is here in full because the third column — what the original actually reports — is the answer to a question a reader of any reproduction should ask, and because "no quantitative figure" is itself a finding that deserves to be checkable row by row.

Section references follow arXiv:2501.10100v1, which uses Roman-numeral sectioning. v2 renumbered to Arabic and moved IV-C's material into Appendix A.4.1. References to arXiv:2504.16680 follow v1, which is the version we read; it is now at v3 (8 Jan 2026), where §5.1 and Eq. 4–5 keep their numbers but every figure and appendix table has moved — Figure 2 (right) became Figure 3 (right), and the model was renamed RWM-O to RWM-U, which is a re-expansion of the letter ("Offline Robotic World Model" to "Uncertainty-Aware Robotic World Model") rather than a second variant: no version of the paper contains both names. All locations, and the occurrence counts that establish that, are recorded in `results/original_paper_figures.json`.

claim	what it needs	order
Sample efficiency, 6,000,000 against ~200M transactions (§V-E)	Isaac Lab, an RTX-class GPU, the MDPPO-PPO loop, and a PPO baseline run to convergence for the comparison	the reference reports 6,000,000 pretraining transactions and 50 min of RWM training on their hardware; the PPO baseline's 200M is the dominant cost
MDPO-PPO beats SAC and DDPG (§V-E)	the above, plus SAC and DDPG implementations at matched budgets	three policy-learning stacks, each tuned enough that the comparison is fair — the largest engineering item here
Zero-shot hardware transfer (§V-E)	all of the above, plus an NVSim, a safe test area, and the site-level stack	not estimable to compute; the budget constraint is hardware access, not GPU lease
Generality across quadruped, humanoid, manipulation (§V-D)	repeated state-action data from a humanoid and a manipulator, which means Isaac Lab and a policy in each environment to generate it — the released CSV is one robot on one terrain	our data-generation run per morphology, plus one world-model training run each at our 4k-2.5 scale; the model training is the cheap half and the data is not
Offline MRL on real robots (2004-2008v1) §5	a real robot, a logged dataset from it, and the offline MRL loop	not estimable to compute; hardware access again, and a claim the follow-up itself states as prospective
Whether the penalty improves the learned policy (2004-2008v1) §5	Isaac Lab, the MDPPO-PPO loop, and at minimum an ablation with the penalty weight at zero	one policy-learning stack; the cheapest of the four, and the one that would leave §11's open question about what the misalignment costs
Best MDP, RSSM, transformer baseline (§V-C)	no simulator needed — but the line release ships only the RSSM variant, so all three baselines would have to be implemented	comparable to our own modified 4k-2.5 of CPU training per architecture, times three, if run at our data budget
M/2, N/8 optimal (§V-C)	no simulator needed; a sweep over M and N at our data budget	our 31 runs took 48.2 h on two cores; a modest sweep is a small multiple of that

F Every pre-registered rule, its lead time and its verdict

§8's argument rests on decision rules committed to git before the data that tested them, and the body names those rules by identifier. An identifier with no table behind it is either decoration or an instruction to open a 430 KB ledger, so here is the table. It is generated from `FINDINGS_LEDGER.md` and `results/appendix_g_rules.json`; nothing in it is typed.

Lead time is the rule's commit timestamp subtracted from the commit that first held the data it tested, resolved by commit *subject* rather than by hash — the history was rewritten once and hashes did not survive it, while subjects did. Positive means the rule was in git before the data existed. This is the same computation Figure 4 plots.

rule	what it governs	commit	lead time	tested by	verdict
R-16	The Arm A / Arm B comparison	84ff019 Step 5: pre-register the decision rule before launching any main run	+13.3 h	first main-run data	SETTLED — rule pre-registered
R-22	Whether episode difficulty biases the A/B comparison	0648a32 Pre-register the Task 0 difficulty-bias rule, and the two-arena convention	+5 min	M-16 re-evaluated	RESOLVED — branch 1, 4c not run
R-23	The 10,000-iteration comparison	4f230af 5.1: pre-register M-23, the long horizon decision rule	+2 min	10k runs launched	RESOLVED — reproduces at long horizon
R-43	The ensemble-5 replication	b17f1a5 PRE-REGISTER the ensemble-5 replication rule, before the runs exist	+13.3 h	en5 result committed	DONES NOT GENERALISE
R-44	The trunk-sharing mechanism	81a4077 PRE-REGISTER M-44 and M-45, with the power check M-43 was committed without	+6.4 h	R2 result committed	MECHANISM SUPPORTED
R-45	The within-trajectory control	81a4077 PRE-REGISTER M-44 and M-45, with the power check M-43 was committed without	+4.4 h	A2 result committed	SUPPORTED
R-49	Pre-registered: does trunk-sharing survive capacity matching?	—	+9.4 h	results/m0_capacity_matched.json	UNDER-POWERED — focuses the matched ensemble by less than the MDE
R-50	Pre-registered: is the sigma collapse driven by the objective, on data whose noise is known?	—	+5 min	results/c5_synthetic_sigma.json	OBJECTIVE-DRIVEN
R-51	Pre-registered: does the ranking claim survive more than one free adversary?	—	+12 min	results/c7_free_baseline.json	SURVIVES entry-oss ONLY
R-52	M-51 named a quantity that does not exist, and what replaces it	—	+9 min	results/c7_free_baseline.json	SURVIVES entry-oss ONLY
R-62	Pre-registered: does any headline verdict change when the bootstrap resamples episodes rather than 400-step trajectories?	—	+5.1 h	results/m02_episode_clustering.json	NO MOVE
R-63	Pre-registered: is the h=1 coverage failure uniform across the 45 state dimensions, or carried by a few?	—	+5.1 h	results/m01_per_dimension_coverage.json	UNIFORM
R-64	Pre-registered: does the horizon-escaped power increase from shorter evaluation units change any verdict at h ≤ 128?	—	+6.0 h	results/m01_short_multi.json	MOVES, at one cell
R-65	Pre-registered: is the Gaussian nominal of 68.27% defensible, or is the error distribution heavy-tailed?	—	+5.1 h	results/m01_gaussian_nominal.json	GAUSSIAN NOMINAL ADEQUATE
S-12	"Task 3's duplication rule was pre-registered"	3ee9d07 Task 3: the duplication control confirms R-47's mechanism and refines its statistic	2.9 h	control runs finished 01-30-21	RETRACTED

15 rules, 15 with a computed lead time, of which 14 are positive and 1 negative. **The negative one is kept deliberately.** S-12 withdraws the claim that the Task 3 duplication rule was pre-registered; the control runs had finished before any threshold reached git. A table that dropped it would be asserting exactly what the ledger retracts.

M-52 is in the table and that is deliberate. It is the one mid-flight amendment to a pre-registration in this project: M-51 named a baseline that does not exist in the artifact it named — the residual on the last teacher-forced step of the history window, which the rollout helper never computes because it copies the history rather than predicting it — and M-52 names the replacement, committed before the replacement's statistic was computed. A table of pre-registrations that omitted the one amendment would be a highlights reel. It was omitted: the selector matched on entry TITLES, and M-52's title does not contain the word, so the row a sceptical reader most wants was silently absent. Entries are selected by their **Status** line as well now, and the count is asserted against the same set **scripts/ledger_check.py** reports.

What each rule says, in its own committed words is in the supplementary material, as docs/APPENDIX_G_RULES.md — every rule's text unabridged, generated from the ledger by the same script that generates this table. Quoting all 15 in full here would add pages to an appendix whose job is to be checkable at a glance, and quoting them in part would ship quotations ending mid-sentence. The table is the claim; the supplementary is the evidence.

G The variance-state arithmetic behind §7.5

§7.5 states the conclusion. This is the arithmetic, kept out of the body because the numbered claim it once supported is retracted (S-19) and because a forensic case the section then defuses with the author's own reply is not what a reader needs in the body of a reproduction.

The collapse rate is a clock. Fitting it across our runs and extrapolating to the released checkpoint's σ state implies **153,270** optimisation steps at the configured learning rate. The refit from our 10,000-iteration runs gives 158,319 and 158,003, spreading 3.3% across the three fits — a linear extrapolation validated over a fourfold extension.

The released configuration says 500 iterations. The paper says 2,500. The checkpoint is tagged 5,000. A second, independent parameter on a slower gradient path implies the same order. And under **gaussian_nll** the implied count is *negative*, which identifies the branch the checkpoint was trained with.

What this extrapolation assumes, and what would falsify it. It assumes constant-rate Adam at the configured learning rate from the released initialisation. Five things would break it, and they are not equally plausible:

assumption	if violated	ruled out by the second parameter?
no learning-rate schedule	a decaying schedule inflates the implied count; a warm-up deflates it	partly — <code>min_logstd</code> and <code>log_delta_logstd</code> travel at rates differing by about 5.2x, and a uniform schedule scales both, so a schedule alone cannot reconcile them without also changing their ratio
<code>log_delta_logstd</code> initialised as released	a different initialisation moves the origin of the fit and rescales the count linearly	no — this is the weakest point of the argument
no warm start from an earlier checkpoint	a warm start makes the count a lower bound on total optimisation, not an estimate of one run	partly — the reference does not clip in the world-model path (X-08), so this is ruled out by source rather than by measurement
no gradient clipping in this path	clipping would slow the collapse and inflate the implied count	partly — same ratio argument as the schedule
bound-less weight as configured	a different weight scales the rate directly	

So the defensible claim is narrower than "cannot have come from the released recipe": **no constant-rate run from the released initialisation at the configured learning rate reaches this checkpoint's variance state in 500, 2,500 or 5,000 iterations.** A warm start or a different initialisation would explain the gap without any inconsistency, and we cannot exclude either.

What the author says. We wrote to the first author on 21 August 2026 asking exactly this. He replied the same day; the exchange is reproduced in full, anonymised, in the supplementary material (SUPPLEMENTARY_CORRESPONDENCE.md): the released `max_iterations: 500` is "a typo"; his recollection is 5,000 iterations, "as I always did"; he does not recall how the checkpoint was obtained; and — the part

that matters most — "the checkpoint was released after a few iterations of the repo than the setup I used for the submission."

That last point reframes this section. The extrapolation above assumes the *released* initialisation and the *released* learning rate. If the repository drifted between the training run and the release, those are not necessarily the values that produced the checkpoint — and a changed `log_delta_logstd` initialisation is precisely the assumption the table above cannot rule out.

So the finding is not that the release is internally inconsistent. It is that **the released artifacts do not reproduce the released checkpoint's variance state, and the author's account is that the released repository is not the one that trained it.** That is a documentation gap between a release and a run — common, worth recording, and much less interesting than an inconsistency. We report the arithmetic because it is what let us detect the gap at all, not as a charge against the work.

H The two nRMSE aggregations, and the comparison one of them inverted

§3.1 states that this paper uses **form 1**: pool the per-dimension mean squared errors across the 45 state dimensions, then divide by the pooled scale — a ratio of means. **Form 2** is the mean of per-dimension ratios.

The two differ because the state dimensions differ in scale by orders of magnitude. Form 2 gives whichever dimension has the smallest denominator unbounded leverage over the aggregate, and a dimension that is nearly constant in the training episodes has a very small denominator. Form 1 has no such lever: a dimension contributes in proportion to its share of the total squared error.

Why this appendix exists rather than a sentence. The choice between the two once inverted a published-model comparison in this project's own history — a result that favoured us under one aggregation and did not under the other. The claim was withdrawn on our own evidence and is kept in the record (FINDINGS_LEDGER.md). Form 2 figures appear nowhere in the body; they are retained in `results/step4_0a_results.json` for continuity with figures this project published before the inversion was found, and are reported here so that continuity does not require trusting a deleted number.

The definition in force throughout the paper is the one §3.1 gives. Nothing in §5, §6 or §7 uses form 2.